

# A Study on Mathematical Logic

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Feb 2026

| Date | Contents                                   | Notes                              |
|------|--------------------------------------------|------------------------------------|
| 3/4  | Intro of Prop. Logic, Structural Induction | <del>1, 2, 3</del>                 |
| 3/11 | Unique Readability, Recursion, Semantics   | <del>4, 5, 6</del>                 |
| 3/18 | Truth tables                               | 7, hw1,2                           |
| 3/25 | Expressive Completeness                    | 8, hw3                             |
| 4/1  | Formal proofs,                             | 9                                  |
| 4/8  | Soundness, Completeness                    | <del>10, 11, 12</del> , hw4        |
| 4/15 | Intro of First Order Logic(FOL), Semantics | <del>13, 14, 15, 16</del> , hw5, 6 |
| 4/22 | Loose ends, Definable sets                 | <del>17, 18, 19, 20</del>          |
| 4/29 | Morphisms                                  | <del>21, 22</del> , hw7            |
| 5/6  | The Back and Forth Method                  | <del>23</del> , hw8                |
| 5/13 | Elementary Substructures                   | <del>24</del>                      |
| 5/20 | Lowenheim-Skolem Theorem                   | <del>25, 26</del>                  |
| 5/27 | Proof of FOL                               | <del>27, 28</del> , hw9            |
| 6/3  | Soundness and Completeness of FOL          | <del>29, 30, 31</del> , hw10       |
| 6/10 | Compactness                                | <del>32, 33, 34, 35</del>          |
| 6/17 | Review & Seminar                           | -                                  |

## 1. Introduction to Mathematical Logic

**Observation:** Valid mathematical reasoning(i.e. proofs) can take many different forms.

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\*Adapted from Patrick Lutz's Math 125A lecture notes at UC Berkeley.

**Key question:** Is it possible to write a computer program which recognizes all valid mathematical arguments? (i.e. can checking the correctness of a proof be turned into a completely mechanical process?)

## 1.1 A brief and incomplete history

Some attempts to formalize mathematical reasoning:

| Period(Approx.)  | Main Development                                                  | Key Figures                         |
|------------------|-------------------------------------------------------------------|-------------------------------------|
| 300 BC           | Axioms (geometry)                                                 | Euclid                              |
| 300 BC – 1800 AD | Syllogisms, Laws of Thought                                       | Aristotle, Vyasa, Avicenna, Leibniz |
| 1850             | Algebraic view of logic                                           | Boole                               |
| 1800 – 1900      | Axioms for $\mathbb{R}$ and $\mathbb{N}$                          | Cauchy, Weierstrass, Dedekind       |
| 1880 – 1940      | Attempt to ground mathematics in logic / set theory               | Frege, Russell, Hilbert             |
| Modern           | Lean proof assistant: Software for formalizing & verifying proofs | Computer Scientists, Mathematicians |

## 1.2 A modern answer

**Key question:** Is it possible to write a computer program which recognizes all valid mathematical arguments? (i.e. can checking the correctness of a proof be turned into a completely mechanical process?)

**Modern approach ( $\geq 1900$ ):**

1. Split mathematical reasoning into 2 parts:
  - (Purely logical reasoning): Reasoning which is purely about logic and does not depend on the specific structures/objects (e.g.,  $\mathbb{N}$ ,  $\mathbb{R}$ ) being reasoned about.
  - (Axioms for specific structures): Axioms that apply only to some specific structure.
2. Find one specific structure and list of axioms (set theory) in which all known math can be developed.

Key results of the modern approach:

- **Logic** It is possible to completely(100%) formalize all valid “purely logical” reasoning.
- **Set Theory** It is possible to reconstruct all known math in one structure / list of axioms.
- **Incompleteness** No computable list of axioms can capture all mathematical truths.

### 1.3 What we cover

1. **Propositional logic:** Logical reasoning about how statements can be combined with logical symbols (connectives) like “and,” “or,” “implies,” etc.  
**E.g.**  $A \rightarrow B$  is equivalent to  $\neg B \rightarrow \neg A$  ( $\neg$  = “not”)
2. **First order logic:** All valid logical reasoning in mathematics
3. **Model theory:** Mathematical applications of the ideas developed for (1) & (2)

### 1.4 A common confusion

In this note, we will give a formal definition of “mathematical proof” and then give informal proofs involving this definition.

**Question:** Why is this okay? How can we prove things if we don’t yet understand formally what a proof is?

**A helpful analogy:**

|                                                                 |                   |                                              |                   |                                             |
|-----------------------------------------------------------------|-------------------|----------------------------------------------|-------------------|---------------------------------------------|
| Informal/intuitive<br>ideas about rates<br>of change/volume/etc | $\longrightarrow$ | Calculus <sup>1</sup><br>formalizes<br>these | $\longrightarrow$ | Proofs about<br>these intuitive<br>concepts |
|-----------------------------------------------------------------|-------------------|----------------------------------------------|-------------------|---------------------------------------------|

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<sup>1</sup>A mathematical model of a pre-existing, real concept

*“The formal definition of ‘mathematical proof’ we give in this note is a **mathematical model** of the intuitive concept of proof, **not a replacement** of it.”*

## 1.5 A puzzle for understanding propositional logic

One or more of your roommates has eaten the food you put in the fridge.

Your roommates are: Alice, Bob, Claire, David, Emily

You know:

1. If Alice ate your food, so did Bob.
2. At least one of David & Emily ate your food.
3. Either Bob or Claire ate your food, but not both.
4. Either both Claire and David ate your food, or neither did.
5. If Emily ate your food, so did Alice and David.

[Question] Who ate your food?

[Solution]

**Claim:** Emily is innocent.

**Proof:** Suppose Emily ate your food.

- Then Alice and David ate your food (by (5)).
- Then Bob ate your food (by (1)).
- Then Claire did not eat your food (by (3)).
- Then David did not eat your food (by (4)).

Contradiction. (Because of David)

Therefore, Emily is innocent.

- Emily is innocent  $\Rightarrow$  David ate your food (by (2)).
- David ate your food  $\Rightarrow$  Claire ate your food (by (4)).

- Claire ate your food  $\Rightarrow$  Bob is innocent (by (3)).
- Bob is innocent  $\Rightarrow$  Alice is innocent (by (1)).

**Answer:** Claire & David ate your food.

**Observation:** The solution did not depend on any facts about food, fridges, people, etc.

More precisely: If  $A, B, C, D, E$  are any statements at all and we know:

1.  $A \Rightarrow B$
2.  $D$  or  $E$
3.  $(B$  or  $C)$  and not  $(B$  and  $C)$
4.  $(C$  and  $D)$  or not  $(C$  and not  $D)$
5.  $E \Rightarrow (A$  and  $D)$

then we can conclude  $C$  and  $D$  are true and  $A, B, E$  are false.

Propositional logic is about deductions like this which depend only on the meaning of words like “and”, “or”, and “not”.

## 2. Propositional logic

### 2.1 Informal language of Propositional Logic

**Goal:** Understand reasoning that depends only on the logical structure of statements & not on their content.

**Example:**

- If ketchup is a vegetable then it is nutritious.  
Ketchup is a vegetable.  
Therefore, ketchup is nutritious.  
**Valid because of the logical form.**
- John goes to elementary school.  
Therefore, John is a child.  
**Valid because of the content.**

**Idea:** Replace statements with letters and just record the logical structure.

**Example:**

$$A = \text{“Ketchup is a vegetable”}, \quad B = \text{“Ketchup is nutritious”}$$

If ketchup is a vegetable then it is nutritious:  $A \rightarrow B$

**Propositional logic:** A formal language in which to express the logical structure of statements.

**Propositional symbols:**  $A, B, C, \dots$  and  $A_0, A_1, A_2, \dots$

**Logical symbols:**

$$\wedge \text{ (and)}, \quad \vee \text{ (or)}, \quad \rightarrow \text{ (implies)}, \quad \neg \text{ (not)}, \quad \top \text{ (true)}, \quad \perp \text{ (false)}$$

**Example:** Either ketchup is a vegetable or it is not nutritious.

$$A \vee (\neg B)$$

### 2.1.1 Some subtleties

1. Many different words can have the same logical meaning:

If  $A$  then  $B$ .     $A$  only if  $B$ .     $A$  implies  $B$ .     $B$  follows from  $A$ .

All mean  $A \rightarrow B$ .

2. The same word can be used in multiple different ways.

$A$  or  $B$  can mean: (i)  $A$  or  $B$  or both.    (ii)  $A$  or  $B$ , but not both.

Sometimes “and” even means “or”.

3. Other issues:

Is “Linda is unhappy” the same as “It is not the case that Linda is happy”?

### 2.1.2 Conventions

- $A \wedge B$ :  $A$  and  $B$  are both true
- $A \vee B$ :  $A$  is true or  $B$  is true or both
- $\neg A$ :  $A$  is not true
- $A \rightarrow B$ : If  $A$  is true then  $B$  is true  
(= either  $A$  is false or  $B$  is true (or both))

#### Examples:

$A$  = “Ketchup is a vegetable”     $B$  = “Ketchup is nutritious”  
 $C$  = “Bread is a vegetable”     $D$  = “Patrick likes bread”

- “Ketchup and bread are both vegetables”     $A \wedge C$
- “Patrick likes bread only if it’s a vegetable”     $D \rightarrow C$
- “Ketchup is not un-nutritious”     $\neg(\neg B)$

## 2.2 Formal language of propositional logic

Before formally defining the language of propositional logic, we need to review a couple things.

**Def:** Given a set  $A^2$ , the set of strings over  $A$ , denoted  $A^*$ , is the set of finite sequences of elements of  $A$ .

#### Example:

$$A = \{0, 1\}, \quad A^* = \{\emptyset, 0, 1, 00, 01, 10, 11, 000, 001, \dots\}$$

**Notation:** Given strings  $\sigma, \tau \in A^*$ :

1.  $|\sigma|$  denotes the length of  $\sigma$ .
2.  $\sigma + \tau$  and  $\sigma \frown \tau$  denote the concatenation of  $\sigma$  and  $\tau$ .

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<sup>2</sup>sometimes called the “alphabet”

**Examples:**

$$\begin{aligned} |01011| &= 5, & |000| &= 3 \\ 011 + 111 &= 011111, & 00 + 111 &= 00111 \end{aligned}$$

**Def:** The alphabet of propositional logic is the set consisting of:

1. Infinitely many propositional symbols:  $A_0, A_1, A_2, \dots$
2. Logical symbols:  $\wedge, \vee, \rightarrow, \neg, \top, \perp$
3. Parentheses:  $(, )$

**Def:** The language of propositional logic is the smallest set  $L$  of strings over the alphabet of propositional logic such that:

- ①  $L$  contains each propositional symbol (i.e. the string consisting of a single propositional symbol).
- ②  $L$  contains  $\top$  and  $\perp$ .
- ③ If  $\varphi, \psi \in L$ , then all of the following are in  $L$ :

$$(\varphi \wedge \psi), \quad (\varphi \vee \psi), \quad (\neg\varphi), \quad (\varphi \rightarrow \psi)$$

Technically,  $(\neg\varphi)$  is the concatenation of the strings “(” +  $\neg$  + “ $\varphi$ ” + “)”.

**Comment:** Elements of the language of propositional logic are often referred to as **formulas** or **well-formed formulas**.

**Example:**

$$((A_0 \rightarrow A_1) \vee (\neg(\perp \wedge A_6)))$$

is a well-formed formula.

- $A_0, A_1, A_6, \perp \in L$  by (1), (2)
- $(A_0 \rightarrow A_1), (\perp \wedge A_6) \in L$  by (3)
- $(\neg(\perp \wedge A_6)) \in L$  by (3)
- $((A_0 \rightarrow A_1) \vee (\neg(\perp \wedge A_6))) \in L$  by (3)

**Non-example:**  $(A_0 \rightarrow \wedge \vee ((A_1$  is not a well-formed formula.



## 2.3 The language of propositional logic

**Alphabet of propositional logic:**

- Propositional symbols:  $A_0, A_1, A_2, \dots$
- Logical symbols:  $\wedge, \vee, \rightarrow, \neg, \top, \perp$
- Parentheses:  $(, )$

**Language of propositional logic:** The **smallest** set  $L$  of strings over the alphabet above such that:

1. For each  $n$ ,  $\langle A_n \rangle \in L$ .
2.  $\langle \top \rangle \in L$  and  $\langle \perp \rangle \in L$ .
3. If  $\varphi, \psi \in L$ , then

$$(\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\neg \varphi) \in L.$$

**Questions:**

1. What do we mean by “ $L$  is the smallest set s.t. ...”?
2. How do we know such a set exists?
3. How should you think about  $L$ ?

### 2.3.1 $L$ is well-defined

**Prop.** There is a unique set of strings  $L$  such that:

1. For each  $n$ ,  $\langle A_n \rangle \in L$ .
2.  $\langle \top \rangle \in L$  and  $\langle \perp \rangle \in L$ .
3. For all  $\varphi, \psi \in L$ ,

$$(\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\neg \varphi) \in L.$$

and moreover, for each set of strings  $M$  which has (1)–(3),  $L \subseteq M$ . (This is what we mean by  $L$  being the smallest)

**Proof idea:**

- **Existence:** Let

$$L = \bigcap \{ M \mid M \text{ has listed properties (1)–(3)} \}.$$

Check  $L$  has (1)–(3).

1. For each  $n$ , for all  $M$  with listed properties,  $\langle A_n \rangle \in M \Rightarrow \langle A_n \rangle \in L$ .
2. For all  $M$  with listed properties,  $\langle T \rangle, \langle \perp \rangle \in M \Rightarrow \langle T \rangle, \langle \perp \rangle \in L$ .
3. If  $\varphi, \psi \in L$ ,  
 $\Rightarrow$  for all  $M$  with listed properties,  $\varphi, \psi \in M$   
 $\Rightarrow$  for all  $M$  with listed properties,  $(\varphi \wedge \psi), (\varphi \vee \psi), \dots \in M$   
 $\Rightarrow (\varphi \wedge \psi), (\varphi \vee \psi), \dots \in L$

- **Smallest:** If  $M$  has listed properties, then  $L \subseteq M$ . by definition of  $L$ .
- **Uniqueness:** If  $L$  and  $L'$  both satisfy conclusion of proposition then  $L \subseteq L'$  and  $L' \subseteq L$ , so  $L = L'$ .

(Summary) If the property  $P$  is preserved under arbitrary intersections, then the intersection of all sets satisfying  $P$  also satisfies  $P$  and is contained in each of them, hence it is the smallest such set; moreover, if two minimal sets satisfied  $P$ , each would be contained in the other, so they must be equal.

## 2.4 How to think about $L$ ?

**Common reaction:** The informal explanation of propositional logic looked simple, but the formal definition looks complicated. (sigh)

**Idea:** Some strings of symbols “make sense” as formulas, and some do not.

**Examples:**

- $((A_0 \wedge A_1) \rightarrow \top)$  makes sense
- $A_0 \wedge \rightarrow$  does not make sense
- $A_0 \wedge A_1 \rightarrow A_2$  does not make sense (ambiguous)

We want a mathematically precise definition of the set of strings that “make sense” so that we can prove things about it.

**Approach:** Define “makes sense” inductively:

1.  $A_n, \top, \perp$  all “make sense”.
2. If  $\varphi, \psi$  “make sense”, then so do  $(\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\neg\varphi)$ .
3. No other strings “make sense”.

## 2.5 How to prove things about $L$

**Observation:** Every proof about well-formed formulas uses the same basic method: **structural induction**.

We will first give an example of this proof technique and then describe it more formally.

**Prop.** Every well-formed formula has the same number of left and right parentheses.

**Proof idea:**

- Let  $L$  be the set of well-formed formulas.
- Let  $M^3$  be the set of strings over the alphabet of propositional logic which have the same number of left and right parentheses.

**Key idea:** Enough to show

- ① For all  $n, \langle A_n \rangle \in M$ .
- ②  $\langle \top \rangle \in M$  and  $\langle \perp \rangle \in M$ .
- ③ For all  $\varphi, \psi \in M$ ,

$$(\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\neg\varphi) \in M.$$

If we can show this, then by definition of  $L$  we have  $L \subseteq M$ , so every string in  $L$  has the property.

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<sup>3</sup>Idea: Let  $M$  be a set with property  $P$ . If  $L \subseteq M$ , then  $L$  also satisfies  $P$ .

**Proof sketch:**

1.  $\langle A_n \rangle$  has no left or right parentheses.
2.  $\langle \top \rangle$  and  $\langle \perp \rangle$  have no left and right parentheses.
3. Suppose  $\varphi, \psi \in M$ . Let

$$n_\varphi = \#(\text{left parentheses in } \varphi), \quad m_\varphi = \#(\text{right parentheses in } \varphi),$$

and similarly  $n_\psi, m_\psi$ . Since  $\varphi, \psi \in M$ , we have  $n_\varphi = m_\varphi$  and  $n_\psi = m_\psi$ .

For example, in  $(\varphi \wedge \psi)$ :

$$\# \text{left} = n_\varphi + n_\psi + 1, \quad \# \text{right} = m_\varphi + m_\psi + 1,$$

which are equal, so  $(\varphi \wedge \psi) \in M$ . The other cases are similar.

## 2.6 Structural induction

**General pattern:** To prove that all well-formed formulas have some property  $P$ , it is enough to prove:

1. For each  $n$ ,  $\langle A_n \rangle$  has property  $P$ .
2.  $\langle \top \rangle$  and  $\langle \perp \rangle$  have property  $P$ .
3. For all  $\varphi, \psi$  with property  $P$ ,

$$(\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\neg \varphi)$$

all have property  $P$ .

**Why does this work?** Let  $L$  be the set of well-formed formulas and let  $M$  be the set of strings with property  $P$ . If (1)–(3) hold, then by definition of  $L$  we have  $L \subseteq M$ , so every string in  $L$  has property  $P$ .

### Another example

**Proposition:** Every well-formed formula contains either some propositional symbol, or  $\top$ , or  $\perp$ .

**Proof (by structural induction):**

1. For each  $n$ ,  $\langle A_n \rangle$  contains a propositional symbol.
2.  $\langle \top \rangle$  contains  $\top$ , and  $\langle \perp \rangle$  contains  $\perp$ .
3. Suppose  $\varphi$  and  $\psi$  both contain either a propositional symbol, or  $\top$ , or  $\perp$ . Then  $(\varphi \wedge \psi)$  contains one of these because  $\varphi$  does (and similarly for  $\psi$ ). The same argument works for  $(\varphi \vee \psi)$ ,  $(\varphi \rightarrow \psi)$ , and  $(\neg \varphi)$ .

## 2.7 More about structural induction

**Structural induction works in many other settings besides propositional logic.**

**Roughly:** Any time you define a set  $S$  as the **smallest** set closed under some operations, you can prove that all elements of  $S$  have some property in a way very similar to the method above.

### Brief Summary

1. Language of propositional logic = set of formulas = set of well-formed formulas.
2.  $\langle A_n \rangle$ ,  $\langle \top \rangle$ ,  $\langle \perp \rangle$  are sometimes called **atomic formulas**.
3. Propositional logic = Sentential logic = Truth-functional logic.
4. Technically, a formula like  $(A_0 \rightarrow (\neg A_1))$  is shorthand for a finite sequence of symbols, e.g.  $(, A_0, \rightarrow, (, \neg, A_1, ), )$  (a finite sequence of length 8).

### 3. The Unique Readability Theorem

**Main goal for today:** Show that it is always possible to parse well-formed formulas (i.e. show that we used enough parentheses)

Formally, we will state and prove the **unique readability theorem**. To understand what this theorem says and why it's useful, it's helpful to consider a language for which it fails.

**Def (weirdly-formed formulas):** The set of weirdly-formed formulas is the smallest set of strings,  $L'$ , such that:

1. Each atomic formula is in  $L'$  (i.e.  $\langle A_n \rangle, \langle \top \rangle, \langle \perp \rangle$ ).
2. For all  $\varphi, \psi \in L'$ ,  $\varphi \wedge \psi, \varphi \vee \psi, \neg \varphi, \varphi \rightarrow \psi$  are all in  $L'$ .

**Comment:** Like well-formed formulas, but no parentheses.

**Examples:**

$$A_0, \quad A_0 \wedge A_1 \rightarrow \neg A_2, \quad \top \vee \perp \vee A_3, \quad \neg \neg \neg A_4$$

**Problem:** Should  $A_0 \vee A_1 \wedge A_2$  be interpreted as

$$A_0 \vee (A_1 \wedge A_2) \quad \text{or} \quad (A_0 \vee A_1) \wedge A_2?$$

This can make a big difference!

**Example meaning:**

$A_0$  = "You'll drink coffee",  $A_1$  = "You'll drink tea",  $A_2$  = "You'll fight a grizzly bear".

#### 3.1 Intro

**Thm (unique readability):** Every well-formed formula has **exactly one** of the following forms:

1.  $\langle A_n \rangle$  for some  $n \in \mathbb{N}$
2.  $\langle \top \rangle$
3.  $\langle \perp \rangle$
4.  $(\varphi \wedge \psi)$  for some well-formed formulas  $\varphi, \psi$

5.  $(\varphi \vee \psi)$  for some well-formed formulas  $\varphi, \psi$
6.  $(\varphi \rightarrow \psi)$  for some well-formed formulas  $\varphi, \psi$
7.  $(\neg\varphi)$  for some well-formed formula  $\varphi$

Moreover, in cases (4)–(6),  $\varphi$  and  $\psi$  are unique, and in case (7),  $\varphi$  is unique.

**Comment:** This fails for “weirdly-formed formulas”. For example,

$$A_0 \vee A_1 \wedge A_2$$

can be read both as  $\varphi_1 \vee \varphi_2$  and as  $\psi_1 \wedge \psi_2$ , where

$$\varphi_1 = A_0, \quad \varphi_2 = A_1 \wedge A_2, \quad \psi_1 = A_0 \vee A_1, \quad \psi_2 = A_2.$$

### 3.2 Another perspective (main connective)

The unique readability theorem implies that every well-formed formula consists of either:

- an atomic formula, or
- a logical symbol connecting one or two subformulas.

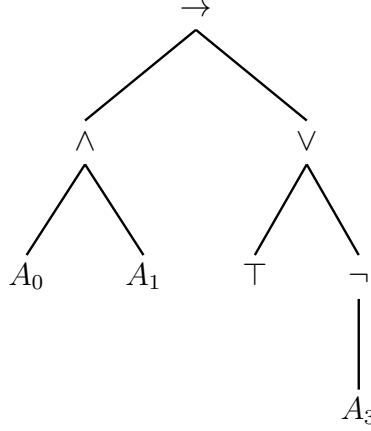
Furthermore, this composition is unique (sometimes people say the formula has a unique **main connective**).

Applying this recursively, we can convert any well-formed formula into a tree (an **abstract syntax tree**).

**Example:**

$$((A_0 \wedge A_1) \rightarrow (\top \vee (\neg A_3)))$$

(main connective:  $\rightarrow$ )



### 3.3 Useful lemmas

Before proving the unique readability theorem, we will prove some lemmas that will help in the proof.

**Lemma 3.1:** Every well-formed formula has one of the following forms:

$$\langle A_n \rangle, \langle \top \rangle, \langle \perp \rangle, (\varphi \wedge \psi), (\varphi \vee \psi), (\varphi \rightarrow \psi), (\neg \varphi).$$

(Like unique readability, but with no uniqueness requirement.)

**Proof idea:** By structural induction (base case: atomic formulas; inductive case: close under the constructors).

**Proof:** By structural induction.

- Base case:  $\langle A_n \rangle$ ,  $\langle \top \rangle$ ,  $\langle \perp \rangle$  all have this property (and are wffs).
- Inductive case: Assume  $\varphi, \psi$  are wffs with this property. Then all of  $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ ,  $(\varphi \rightarrow \psi)$ ,  $(\neg \varphi)$  are also wffs with this property.

**Lemma 3.2:** No well-formed formula starts with the symbol  $\neg$ .

**Proof idea:** By the previous lemma, it is enough to check the cases:

- $\langle A_n \rangle$  starts with  $A_n$ , not  $\neg$ .
- $\langle \top \rangle$  starts with  $\top$ , not  $\neg$ .
- $\langle \perp \rangle$  starts with  $\perp$ , not  $\neg$ .



- $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ ,  $(\varphi \rightarrow \psi)$  start with “(”, not  $\neg$ .
- $(\neg\varphi)$  starts with “(”, not  $\neg$ .

**Lemma 3.3:** Every well-formed formula contains the same number of left and right parentheses.

**Proof:** Proved in Lecture 3.

### 3.4 Initial segments

**Def:** Given a string  $\sigma = \langle a_0, \dots, a_n \rangle$ , a string  $\tau$  is an **initial segment** of  $\sigma$  if

$$\tau = \emptyset \quad \text{or} \quad \tau = \langle a_0, \dots, a_m \rangle \text{ for some } m \leq n.$$

**Def:**  $\tau$  is a **proper initial segment** of  $\sigma$  if  $\tau$  is an initial segment of  $\sigma$  which is neither empty nor equal to  $\sigma$ .

**Example:** Let  $\sigma = 110101$ .

- $\tau_1 = \emptyset$  is an initial segment, but not a proper initial segment.
- $\tau_2 = 110$  is an initial segment and a proper initial segment.
- $\tau_3 = 101$  is not an initial segment.
- $\tau_4 = 110101$  is an initial segment, but not a proper initial segment.
- $\tau_5 = 11010111$  is not an initial segment.

### 3.5 A key lemma about parentheses

**Lemma 3.4:** If  $\varphi$  is a well-formed formula, then every proper initial segment of  $\varphi$  has strictly more left parentheses than right parentheses.

**Proof idea:** By structural induction.

- Base case:  $\langle A_n \rangle$ ,  $\langle \top \rangle$ ,  $\langle \perp \rangle$  have no proper initial segments.
- Inductive case: Suppose  $\varphi, \psi$  are wffs with this property and consider  $\sigma = (\varphi \wedge \psi)$ . Let  $\tau$  be a proper initial segment of  $\sigma$ . Then  $\tau$  must be one of:

$$(\quad, \quad (\varphi', \quad (\varphi, \quad (\varphi \wedge, \quad (\varphi \wedge \psi', \quad (\varphi \wedge \psi$$

where  $\varphi'$  is a proper initial segment of  $\varphi$  and  $\psi'$  is a proper initial segment of  $\psi$ . Each case can be checked using the induction assumption.

Cases for  $(\varphi \vee \psi)$ ,  $(\varphi \rightarrow \psi)$ , and  $(\neg\varphi)$  are similar.

**Corollary 3.5:** No proper initial segment of a well-formed formula is itself a well-formed formula. (A useful consequence with Lemma 3.3 and Lemma 3.4)

### 3.6. Proof of unique readability

By a previous lemma, it is enough to consider the following cases (We will just check ‘Is it unique?’):

1.  $\langle A_n \rangle$ ,  $\langle \top \rangle$ ,  $\langle \perp \rangle$
2.  $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ , or  $(\varphi \rightarrow \psi)$  for some wffs  $\varphi, \psi$
3.  $(\neg\varphi)$  for some wff  $\varphi$

We will just check one of these cases; the others are similar.

Suppose  $\varphi, \psi$  are wffs and consider  $(\varphi \wedge \psi)$ . We need to show:

1. For all  $n$ ,  $(\varphi \wedge \psi) \neq \langle A_n \rangle$ .
2.  $(\varphi \wedge \psi) \neq \langle \top \rangle$ .
3.  $(\varphi \wedge \psi) \neq \langle \perp \rangle$ .
4. For all wffs  $\varphi', \psi'$ ,  $(\varphi \wedge \psi) \neq (\varphi' \vee \psi')$ .
5. For all wffs  $\varphi', \psi'$ ,  $(\varphi \wedge \psi) \neq (\varphi' \rightarrow \psi')$ .
6. For all wffs  $\varphi'$ ,  $(\varphi \wedge \psi) \neq (\neg\varphi')$ .
7. For all wffs  $\varphi', \psi'$ , if  $(\varphi \wedge \psi) = (\varphi' \wedge \psi')$  then  $\varphi = \varphi'$  and  $\psi = \psi'$ .

**Sketch of (4):** Let  $\varphi', \psi'$  be wffs and suppose for contradiction

$$(\varphi \wedge \psi) = (\varphi' \vee \psi').$$

Compare lengths  $|\varphi|$  and  $|\varphi'|$ :

- If  $|\varphi| < |\varphi'|$ , then  $\varphi$  is a proper initial segment of  $\varphi'$  (contradiction to the corollary 3.5).

- If  $|\varphi'| < |\varphi|$ , then  $\varphi'$  is a proper initial segment of  $\varphi$  (contradiction).
- If  $|\varphi| = |\varphi'|$ , then  $\wedge$  and  $\vee$  occur at the same position, so the strings cannot be equal.

So  $(\varphi \wedge \psi) \neq (\varphi' \vee \psi')$ . The other “different connective” cases are similar.

**Sketch of (6):** Suppose for contradiction  $(\varphi \wedge \psi) = (\neg\varphi')$ . Then the first symbol in  $\varphi$  is  $\neg$ , contradicting the lemma 3.2 that no wff starts with  $\neg$ .

**Sketch of (7):** Suppose  $(\varphi \wedge \psi) = (\varphi' \wedge \psi')$ . By reasoning like in (4), we get  $|\varphi| = |\varphi'|$ , hence  $\varphi = \varphi'$ . Then  $\psi$  and  $\psi'$  begin and end at the same positions, so  $\psi = \psi'$ .

## 4. Structural Recursion

### 4.1 Recursion, informally

**Goal:** Show how to define functions on formulas by recursion on the structure of the formula.

**Informal example:** The **depth** of a formula is defined by:

1. The depth of an atomic formula is 1.
2. The depth of a formula of the form  $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ , or  $(\varphi \rightarrow \psi)$  is

$$\max(\text{depth}(\varphi), \text{depth}(\psi)) + 1.$$

3. The depth of a formula of the form  $(\neg\varphi)$  is

$$\text{depth}(\varphi) + 1.$$

**Example:** Compute  $\text{depth}(((A_0 \vee A_1) \rightarrow \top))$ .

$$\begin{aligned} \text{depth}(((A_0 \vee A_1) \rightarrow \top)) &= \max(\text{depth}((A_0 \vee A_1)), \text{depth}(\top)) + 1 \\ &= \max\left(\max(\text{depth}(A_0), \text{depth}(A_1)) + 1, \text{depth}(\top)\right) + 1 \\ &= \max(\max(1, 1) + 1, 1) + 1 \\ &= 3. \end{aligned}$$

**Question:** Why is this a valid way to define a function?

**Answer:** Essentially follows from **unique readability**.

Soon, we will state a formal theorem that justifies this way of defining functions on the set of wffs.

## 4.2 Structural Recursion

The following theorem justifies this kind of recursive definition.

**Thm (Structural Recursion):** Let  $L$  be the set of well-formed formulas (wffs). Suppose we have a set  $X$  and functions

$$f_A : \{\top, \perp, A_0, A_1, A_2, \dots\} \rightarrow X, \quad f_\wedge : X \times X \rightarrow X, \quad f_\vee : X \times X \rightarrow X, \\ f_\rightarrow : X \times X \rightarrow X, \quad f_\neg : X \rightarrow X.$$

Then there is a **unique** function  $g : L \rightarrow X$  such that:

1. For all  $\varphi \in \{\top, \perp, A_0, A_1, \dots\}$ ,

$$g(\varphi) = f_A(\varphi).$$

2. For all  $\varphi, \psi \in L$ ,

$$g((\varphi \wedge \psi)) = f_\wedge(g(\varphi), g(\psi)), \quad g((\varphi \vee \psi)) = f_\vee(g(\varphi), g(\psi)), \\ g((\varphi \rightarrow \psi)) = f_\rightarrow(g(\varphi), g(\psi)), \quad g((\neg \varphi)) = f_\neg(g(\varphi)).$$

**Example (depth via the theorem):** To justify our definition of depth using this theorem, take:

$$X = \mathbb{N}, \quad f_A(\varphi) = 1, \\ f_\neg(n) = n + 1, \quad f_\wedge(n, m) = f_\vee(n, m) = f_\rightarrow(n, m) = \max(n, m) + 1.$$

## 4.3 Another perspective

Given a recursive definition of a function  $g$  on the set of wffs in terms of  $f_A, f_\wedge, f_\vee, f_\rightarrow, f_\neg$ , the value  $g(\varphi)$  can be evaluated using the **abstract syntax tree** of  $\varphi$ .

**Example:** Evaluate  $\text{depth}(((A_0 \vee A_1) \rightarrow \top))$  via its tree.

$$\begin{array}{c} \rightarrow \quad (3) \\ / \quad \backslash \\ \vee \quad (2) \quad \top \quad (1) \\ / \quad \backslash \\ A_0(1) \quad A_1(1) \end{array}$$

**Procedure:**

- First evaluate the function at the leaves (one by one).
- Then evaluate at internal nodes (one by one).
- The value at the root is the value of  $g(\varphi)$ .

#### 4.4 A stronger form of structural recursion

We would like to use structural recursion to justify the following definition.

**Def:** The set of **subformulas** of a wff  $\varphi$ , denoted  $\text{Subform}(\varphi)$ , is defined by structural recursion as follows:

1. For any atomic formula  $\varphi$ ,

$$\text{Subform}(\varphi) = \{\varphi\}.$$

2. For any wffs  $\varphi$  and  $\psi$ :

$$\text{Subform}((\varphi \wedge \psi)) = \text{Subform}(\varphi) \cup \text{Subform}(\psi) \cup \{(\varphi \wedge \psi)\},$$

$$\text{Subform}((\varphi \vee \psi)) = \text{Subform}(\varphi) \cup \text{Subform}(\psi) \cup \{(\varphi \vee \psi)\},$$

$$\text{Subform}((\varphi \rightarrow \psi)) = \text{Subform}(\varphi) \cup \text{Subform}(\psi) \cup \{(\varphi \rightarrow \psi)\},$$

$$\text{Subform}((\neg\varphi)) = \text{Subform}(\varphi) \cup \{(\neg\varphi)\}.$$

**Examples:**

$$\text{Subform}(\top) = \{\top\}, \quad \text{Subform}(A_0) = \{A_0\},$$

$$\text{Subform}((A_0 \vee A_1)) = \{A_0, A_1, (A_0 \vee A_1)\},$$

$$\text{Subform}(((A_0 \vee A_1) \rightarrow \top)) = \{A_0, A_1, (A_0 \vee A_1), \top, ((A_0 \vee A_1) \rightarrow \top)\}.$$

**Problem:** This doesn't easily fit into our previous framework because, e.g.,  $\text{Subform}((\varphi \wedge \psi))$  depends not just on  $\text{Subform}(\varphi)$  and  $\text{Subform}(\psi)$ , but also on  $\varphi$  and  $\psi$  themselves.

**Solution:** Use a stronger version of structural recursion.

## 4.5 A stronger version of structural recursion

Let  $L$  be the set of wffs.

**Thm (strong structural recursion):** Suppose we have a set  $X$  and functions

$$\begin{aligned} f_A &: \{\top, \perp, A_0, A_1, A_2, \dots\} \rightarrow X, \\ f_\wedge &: L \times L \times X \times X \rightarrow X, \\ f_\vee &: L \times L \times X \times X \rightarrow X, \quad f_\rightarrow : L \times L \times X \times X \rightarrow X, \\ f_\neg &: L \times X \rightarrow X. \end{aligned}$$

Then there is a **unique** function  $g : L \rightarrow X$  such that:

1. For all  $\varphi \in \{\top, \perp, A_0, A_1, \dots\}$ ,

$$g(\varphi) = f_A(\varphi).$$

2. For all  $\varphi, \psi \in L$ ,

$$g((\varphi \wedge \psi)) = f_\wedge(\varphi, \psi, g(\varphi), g(\psi)),$$

$$g((\varphi \vee \psi)) = f_\vee(\varphi, \psi, g(\varphi), g(\psi)),$$

$$g((\varphi \rightarrow \psi)) = f_\rightarrow(\varphi, \psi, g(\varphi), g(\psi)),$$

$$g((\neg\varphi)) = f_\neg(\varphi, g(\varphi)).$$

**Defining Subform using the theorem:** Take

$$X = \mathcal{P}(L) \quad (\text{the powerset of the set of wffs}), \quad f_A(\varphi) = \{\varphi\}.$$

Define:

$$f_\wedge(\varphi, \psi, A, B) = A \cup B \cup \{(\varphi \wedge \psi)\},$$

$$f_\vee(\varphi, \psi, A, B) = A \cup B \cup \{(\varphi \vee \psi)\},$$

$$f_\rightarrow(\varphi, \psi, A, B) = A \cup B \cup \{(\varphi \rightarrow \psi)\},$$

$$f_\neg(\varphi, A) = A \cup \{(\neg\varphi)\}.$$

Then the unique  $g : L \rightarrow \mathcal{P}(L)$  given by the theorem is exactly Subform.

## 5. Semantics

### 5.1 Semantics, informally

Given a formula of propositional logic, if we assign a meaning to each propositional symbol then we can determine if the entire formula is true or false.

**Example:** Let

$$\varphi = ((A_0 \vee A_1) \wedge A_2).$$

If we set

$A_0 = \text{“Berkeley is in CA”}$ ,  $A_1 = \text{“The sky is green”}$ ,  $A_2 = \text{“The ocean is wet”}$ ,

then  $\varphi$  is true.

If instead we set

$A_0 = \text{“The sky is green”}$ ,  $A_1 = \text{“The sky is purple”}$ ,  $A_2 = \text{“The ocean is wet”}$ ,

then  $\varphi$  is false.

We are interested in questions like:

- Which formulas are always true no matter what the propositional symbols mean?
- When does the truth of one formula automatically imply the truth of another?

### 5.2 Tautology / Contradiction / Satisfiable (informal)

**Def:** A formula  $\varphi$  is a:

- **tautology** (a.k.a. **validity**) if every assignment of meanings to the propositional symbols of  $\varphi$  makes  $\varphi$  true.
- **contradiction** if every assignment of meanings to the propositional symbols of  $\varphi$  makes  $\varphi$  false.
- **satisfiable** if there is some assignment of meanings to the propositional symbols of  $\varphi$  which makes  $\varphi$  true.

**Examples:**



- $(A_0 \vee (\neg A_0))$  is a tautology (and satisfiable).
- $(A_0 \vee A_1)$  is satisfiable but not a tautology.
- $(A_0 \wedge (\neg A_0))$  is a contradiction.

### 5.3 Logical implication (informal definition)

**Def:** A set of formulas  $\Gamma$  **logically implies** a formula  $\varphi$ , written  $\Gamma \models \varphi$ , if every assignment of meanings to the propositional symbols in  $\Gamma$  and  $\varphi$  which makes every formula in  $\Gamma$  true also makes  $\varphi$  true.

**Examples:**

1.  $\{A_0, A_1\} \models (A_0 \vee A_2)$  True
2.  $\{A_0, (A_0 \rightarrow A_1)\} \models A_1$  True
3.  $\{A_0, A_1\} \models (A_0 \wedge A_2)$  False

**Comment:**  $\varphi$  is a tautology  $\iff \emptyset \models \varphi$ .

**Convention:**

$\models \varphi$  denotes  $\emptyset \models \varphi$ ,       $\psi \models \varphi$  denotes  $\{\psi\} \models \varphi$ ,

$\psi_1, \psi_2 \models \varphi$  denotes  $\{\psi_1, \psi_2\} \models \varphi$ .

### 5.3 Semantics, formally

**Question:** How can we define concepts like tautology/etc. in a more mathematically precise way?

Also, how can we check if a formula is a tautology/etc.?

**Key observation:** Whether  $\varphi$  is true or not does not depend on the exact meaning assigned to each propositional symbol, just whether it is **true or false**.

**Example:** Let  $\varphi = (A_0 \vee A_1)$ . If

$A_0 = \text{“Berkeley is in CA”}, \quad A_1 = \text{“UCLA is in OR”},$

then  $\varphi$  is true because  $A_0$  is true, not because of the specific content of  $A_0$ .

## 5.4 Truth assignments

Let  $S$  denote the set of propositional symbols:

$$S = \{A_0, A_1, A_2, \dots\}.$$

**Def:** A **truth assignment** is a function

$$v : S \rightarrow \{T, F\}.$$

**Def:** Given a truth assignment  $v : S \rightarrow \{T, F\}$ , define a function

$$\bar{v} : L \rightarrow \{T, F\}$$

(where  $L$  is the set of all wffs) by recursion as follows:

$$\begin{aligned}\bar{v}(A_n) &= v(A_n) \\ \bar{v}(\top) &= T \\ \bar{v}(\perp) &= F \\ \bar{v}((\varphi \wedge \psi)) &= \begin{cases} T & \text{if } \bar{v}(\varphi) = T \text{ and } \bar{v}(\psi) = T \\ F & \text{else} \end{cases} \\ \bar{v}((\varphi \vee \psi)) &= \begin{cases} T & \text{if } \bar{v}(\varphi) = T \text{ or } \bar{v}(\psi) = T \\ F & \text{else} \end{cases} \\ \bar{v}((\varphi \rightarrow \psi)) &= \begin{cases} T & \text{if } \bar{v}(\psi) = T \text{ or } \bar{v}(\varphi) = F \\ F & \text{else} \end{cases} \\ \bar{v}((\neg \varphi)) &= \begin{cases} T & \text{if } \bar{v}(\varphi) = F \\ F & \text{if } \bar{v}(\varphi) = T \end{cases}\end{aligned}$$

**Notation:**

$$v \models \varphi \text{ means } \bar{v}(\varphi) = T, \quad v \models \Gamma \text{ means } \forall \varphi \in \Gamma, \bar{v}(\varphi) = T,$$

where  $\Gamma$  is some set of formulas.

## 5.5 Examples

**Example truth assignment:** Define

$$v(A_n) = \begin{cases} T & \text{if } n \text{ is even} \\ F & \text{if } n \text{ is odd.} \end{cases}$$

Then:

$$\bar{v}(((A_0 \vee A_1) \wedge A_2)) = T, \quad \bar{v}(((A_1 \vee A_3) \wedge A_4)) = F.$$

So:

$$\begin{aligned} v &\models ((A_0 \vee A_1) \wedge A_2), & v &\not\models ((A_1 \vee A_3) \wedge A_4), \\ v &\models \{A_0, \top, ((A_0 \vee A_1) \wedge A_2)\}, & v &\not\models \{((A_0 \vee A_1) \wedge A_2), ((A_1 \vee A_3) \wedge A_4)\}. \end{aligned}$$

## 5.6 Tautology / Contradiction / Satisfiable (formal)

**Def:** A formula  $\varphi$  is a:

- **tautology / validity** if for every truth assignment  $v$ ,  $v \models \varphi$ .
- **contradiction** if for every truth assignment  $v$ ,  $v \not\models \varphi$ .
- **satisfiable** if for some truth assignment  $v$ ,  $v \models \varphi$ .

**Def:** A set of formulas  $\Gamma$  **logically implies** a formula  $\varphi$ , written  $\Gamma \models \varphi$ , if for every truth assignment  $v$ :

$$\text{if } v \models \Gamma \text{ then } v \models \varphi.$$

**Comment:** The symbol  $\models$  is used in two somewhat different ways:

1. To express that a formula/set of formulas is satisfied by a truth assignment.
2. To express that a formula is logically implied by another formula/set of formulas.

## 5.7 A comment about implication

Our formal semantics for  $(\varphi \rightarrow \psi)$  might look a bit odd.

**Recall:**

$$\bar{v}((\varphi \rightarrow \psi)) = \begin{cases} T & \text{if } \bar{v}(\psi) = T \text{ or } \bar{v}(\varphi) = F \\ F & \text{else.} \end{cases}$$

i.e.  $(\varphi \rightarrow \psi)$  is true iff  $\psi$  is true or  $\varphi$  is false.

**Examples:**

- If the sky is green then I am a carrot.    True
- If the sky is green then the sky is blue.    True
- If the sky is blue then the sky is green.    False

This is sometimes called **material implication**.

It is different from the notion of implication used sometimes in natural language, where implication suggests some level of causality.

Material implication, though, is what “implies” usually means in math.

## 6. Truth Tables

### 6.1 Some conventions

We will often deviate from our formal definition of “well-formed formula”:

1. We will sometimes use  $A, B, C, \dots$  as propositional symbols (instead of  $A_0, A_1, \dots$ ).
2. We will assume  $\neg$  “takes precedence” over  $\wedge, \vee, \rightarrow$  and skip some parentheses.  
E.g.  $(\neg A \wedge B)$  means  $((\neg A) \wedge B)$ .
3. We will assume  $\wedge$  and  $\vee$  take precedence over  $\rightarrow$ .  
E.g.  $(A \wedge B \rightarrow C)$  means  $((A \wedge B) \rightarrow C)$ .
4.  $\vee, \wedge, \rightarrow$  “associate to the right”.  
E.g.  $(A \rightarrow B \rightarrow C)$  means  $(A \rightarrow (B \rightarrow C))$ .

5. We will often leave off the outermost parentheses.  
E.g.  $(A \wedge B) \vee C$ .

## 6.2 Truth tables

**Question:** How can we check whether  $\varphi$  is a tautology, whether  $\varphi \models \psi$ , etc.?

**Observation:** To compute  $\bar{v}(\varphi)$  we just need to know the values that  $v$  assigns to the propositional symbols that actually appear in  $\varphi$ .

So to check if  $\varphi$  is a tautology, we just check all truth assignments to the propositional symbols which appear in  $\varphi$ .

**Truth tables:** A systematic way to organize this.

**Example.** Let  $\varphi = ((A \vee B) \rightarrow A)$ .

| $A$ | $B$ | $A \vee B$ | $(A \vee B) \rightarrow A$ |
|-----|-----|------------|----------------------------|
| T   | T   | T          | T                          |
| T   | F   | T          | T                          |
| F   | T   | T          | F                          |
| F   | F   | F          | T                          |

The third row says  $\varphi$  is **not** a tautology.

**How to build a truth table (systematically).**

- First list all propositional symbols in  $\varphi$ .
- Then list all other subformulas of  $\varphi$  so that each one occurs after all of its subformulas.
- The last column corresponds to  $\varphi$ .

Each row gives a truth assignment and the truth values it assigns to each subformula of  $\varphi$ .

**Comment:** A truth table for a formula with  $n$  distinct propositional symbols has  $2^n$  rows.

### 6.3 Example & Other Uses

**Pierce's law:**  $((A \rightarrow B) \rightarrow A) \rightarrow A$ .

**Question:** Is Pierce's law a tautology?

Truth table:

| $A$ | $B$ | $A \rightarrow B$ | $(A \rightarrow B) \rightarrow A$ | $((A \rightarrow B) \rightarrow A) \rightarrow A$ |
|-----|-----|-------------------|-----------------------------------|---------------------------------------------------|
| T   | T   | T                 | T                                 | T                                                 |
| T   | F   | F                 | T                                 | T                                                 |
| F   | T   | T                 | F                                 | T                                                 |
| F   | F   | T                 | F                                 | T                                                 |

Pierce's law **is** a tautology.

**Here's other uses:**

Truth tables can also be used to check:

1. **If  $\varphi$  is a contradiction.**

E.g.  $\varphi = A \wedge \neg B \wedge (A \rightarrow B)$ .

| $A$ | $B$ | $\neg B$ | $A \rightarrow B$ | $A \wedge \neg B$ | $A \wedge \neg B \wedge (A \rightarrow B)$ |
|-----|-----|----------|-------------------|-------------------|--------------------------------------------|
| T   | T   | F        | T                 | F                 | F                                          |
| T   | F   | T        | F                 | T                 | F                                          |
| F   | T   | F        | T                 | F                 | F                                          |
| F   | F   | T        | T                 | F                 | F                                          |

Last column is all F, so  $\varphi$  is a contradiction.

2. **If  $\varphi$  is satisfiable.**

E.g.  $\varphi = A \wedge (A \rightarrow B)$ .

| $A$ | $B$ | $A \rightarrow B$ | $A \wedge (A \rightarrow B)$ |
|-----|-----|-------------------|------------------------------|
| T   | T   | T                 | T                            |
| T   | F   | F                 | F                            |
| F   | T   | T                 | F                            |
| F   | F   | T                 | F                            |

There is at least one T in the last column  $\Rightarrow$  satisfiable.

3. If  $\varphi \models \psi$  or if  $\Gamma \models \varphi$  (as long as  $\Gamma$  is finite).

E.g.  $\{A \rightarrow B, \neg B\} \models \neg A$ .

| $A$ | $B$ | $A \rightarrow B$ | $\neg B$ | $\neg A$ |
|-----|-----|-------------------|----------|----------|
| T   | T   | T                 | F        | F        |
| T   | F   | F                 | T        | F        |
| F   | T   | T                 | F        | T        |
| F   | F   | T                 | T        | T        |

Look at all rows in which every formula in  $\Gamma$  is true (here:  $A \rightarrow B$  and  $\neg B$  both true). In that row,  $\neg A$  is true.

## 6.4 Logical equivalence

**Def.** Formulas  $\varphi$  and  $\psi$  are **logically equivalent**, denoted  $\varphi \equiv \psi$ , if

$$\varphi \models \psi \quad \text{and} \quad \psi \models \varphi.$$

**Comments.**

1. Sometimes also denoted  $\varphi \models \psi$
- 2.

$$\varphi \equiv \psi$$

$$\iff \forall v, \bar{v}(\varphi) = \bar{v}(\psi)$$

$$\iff (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi) \text{ is a tautology.}$$

Truth tables can also be used to check logical equivalence.

**Example.**  $\neg(A \wedge B) \equiv (\neg A \vee \neg B)$ .

| $A$ | $B$ | $A \wedge B$ | $\neg A$ | $\neg B$ | $\neg(A \wedge B)$ | $\neg A \vee \neg B$ |
|-----|-----|--------------|----------|----------|--------------------|----------------------|
| T   | T   | T            | F        | F        | F                  | F                    |
| T   | F   | F            | F        | T        | T                  | T                    |
| F   | T   | F            | T        | F        | T                  | T                    |
| F   | F   | F            | T        | T        | T                  | T                    |

Last two columns are the same  $\Rightarrow$  logically equivalent.

## 6.5 Some examples of logical equivalence

- **Distributivity:**

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C), \quad A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C).$$

- **De Morgan's laws:**

$$\neg(A \wedge B) \equiv \neg A \vee \neg B, \quad \neg(A \vee B) \equiv \neg A \wedge \neg B.$$

- **Negation:**

$$\neg\neg A \equiv A, \quad \neg(A \rightarrow B) \equiv A \wedge \neg B.$$

- **Contraposition:**

$$A \rightarrow B \equiv \neg B \rightarrow \neg A.$$

- **Exportation:**

$$A \rightarrow (B \rightarrow C) \equiv (A \wedge B) \rightarrow C.$$

- **Other:**

$$A \rightarrow B \equiv \neg A \vee B.$$

## 6.6 Expressive completeness

Our choice of what logical symbols to include in our formal definition of the language of propositional logic was somewhat arbitrary.

**Example:** Enderton includes  $\leftrightarrow$ .

**Question:** How do we know we're not "missing" anything?  
i.e. how do we know every basic logical operation can be expressed in our language?

We will give two examples to illustrate what this means.



### 6.6.1 Example 1

Suppose we had included  $\leftrightarrow$  in our language with semantics given by:

$$\bar{v}(\varphi \leftrightarrow \psi) = \begin{cases} \text{T} & \text{if } \bar{v}(\varphi) = \bar{v}(\psi), \\ \text{F} & \text{else.} \end{cases}$$

It is possible to show:

1. For any formulas  $\varphi, \psi$  in this larger language,

$$\varphi \leftrightarrow \psi \equiv (\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi).$$

2. For any formula  $\varphi$  in this larger language, there is a formula  $\psi$  without the symbol  $\leftrightarrow$  such that  $\varphi \equiv \psi$ .  
(To see why, recursively apply (1).)

**Example.** Let  $R := ((B \wedge C) \leftrightarrow A)$ . Then

$$A \leftrightarrow R \equiv (A \rightarrow R) \wedge (R \rightarrow A)$$

and

$$R \equiv ((B \wedge C) \rightarrow A) \wedge (A \rightarrow (B \wedge C)).$$

So one equivalent formula using only  $\neg, \wedge, \vee, \rightarrow$  is:

$$\left( A \rightarrow (((B \wedge C) \rightarrow A) \wedge (A \rightarrow (B \wedge C))) \right) \wedge \left( (((B \wedge C) \rightarrow A) \wedge (A \rightarrow (B \wedge C))) \rightarrow A \right).$$

### 6.6.2 Example 2

Suppose we had only included  $\wedge$  and  $\vee$  in our language and not  $\neg, \rightarrow, \top, \perp$ .

Then there would be sentences we could not express!

**Key observation:** If  $\varphi$  contains only  $\wedge, \vee$ , and propositional symbols, and if  $v$  is the truth assignment that assigns T to every propositional symbol, then  $\bar{v}(\varphi) = \text{T}$ .

(This can be proved by structural induction.)

**Example.** Let  $\varphi = (A \wedge B) \vee (B \wedge (C \vee A))$  and suppose  $v(A) = v(B) = v(C) = \text{T}$ . Then

$$\bar{v}(A \wedge B) = \text{T}, \quad \bar{v}(C \vee A) = \text{T}, \quad \bar{v}(B \wedge (C \vee A)) = \text{T},$$

so  $\bar{v}(\varphi) = \text{T}$ .

**Consequence:** No formula which contains only  $\wedge, \vee$ , and propositional symbols is equivalent to  $\neg A_0$ .

## 7. Expressive Completeness

### 7.1 Boolean Functions

**Goal:** Show that our language of propositional logic is not missing anything; i.e. it can express any statement we could want it to.

**Obstacle:** How can we even express this idea mathematically?

**Key idea:** Think of a formula as a function from truth assignments to truth values and show that every such function is given by some formula. This function is called a **Boolean function**.

### 7.2 Definition and Examples

**Def.** An  $n$ -ary **Boolean function** is a function

$$f : \{\text{T}, \text{F}\}^n \rightarrow \{\text{T}, \text{F}\}.$$

**Examples.**

1.  $f : \{\text{T}, \text{F}\}^2 \rightarrow \{\text{T}, \text{F}\}$  defined by

$$f(X, Y) = \begin{cases} \text{T} & \text{if } X = Y, \\ \text{F} & \text{else.} \end{cases}$$

(So  $f(\text{T}, \text{F}) = \text{F}$  and  $f(\text{F}, \text{F}) = \text{T}$ , etc.)

2.  $g : \{\text{T}, \text{F}\}^3 \rightarrow \{\text{T}, \text{F}\}$  defined by  $g(X, Y, Z) =$  the majority value of  $X, Y, Z$ .

E.g.  $g(\text{T}, \text{F}, \text{T}) = \text{T}$ ,  $g(\text{F}, \text{F}, \text{F}) = \text{F}$ ,  $g(\text{F}, \text{F}, \text{T}) = \text{F}$ .

**Idea:** Every formula gives a Boolean function.

**Example.** If  $\varphi = (A_0 \wedge A_1) \vee A_2$ , then the corresponding 3-ary Boolean function is

$$f(X, Y, Z) = (X \wedge Y) \vee Z.$$

E.g.  $f(\text{T}, \text{F}, \text{F}) = \text{F}$ ,  $f(\text{F}, \text{F}, \text{T}) = \text{T}$ ,  $f(\text{T}, \text{T}, \text{F}) = \text{T}$ .

**Def.** Given a formula  $\varphi$  and a number  $n$  such that all propositional symbols in  $\varphi$  are contained in  $\{A_0, \dots, A_{n-1}\}$ , define an  $n$ -ary Boolean function  $f_\varphi^n$  by:

$$f_\varphi^n(X_0, \dots, X_{n-1}) = \bar{v}(\varphi),$$

where  $v$  is any truth assignment such that

$$v(A_0) = X_0, \dots, v(A_{n-1}) = X_{n-1}.$$

**Examples.**

$$f_{A_0}^1(X) = X, \quad f_{A_0}^2(X, Y) = X, \quad f_{A_0 \wedge A_1}^2(X, Y) = \begin{cases} \text{T} & \text{if } X = Y = \text{T}, \\ \text{F} & \text{else.} \end{cases}$$

Also,  $f_{A_0 \wedge A_1}^1(X)$  does not make sense ( $n$  is too small).

**Technicality:** Why is  $f_\varphi^n$  well-defined? Why doesn't it depend on the choice of  $v$ ?

**Prop.** Suppose  $\varphi$  is a formula and  $v, \mu$  are truth assignments which agree on every propositional symbol in  $\varphi$ . Then

$$\bar{v}(\varphi) = \bar{\mu}(\varphi).$$

**pf.** By structural induction.

**Base case:**  $\varphi$  is atomic (e.g.  $A_n$ ) or  $\top$  or  $\perp$ . Then the value of  $\bar{v}(\varphi)$  depends only on the value of that symbol, so agreement implies equality.

**Inductive case:** Suppose  $\varphi = (\psi_1 \wedge \psi_2)$ . (Other cases are similar.) Since  $v, \mu$  agree on all propositional symbols in  $\varphi$ , they agree on all propositional symbols in  $\psi_1$  and  $\psi_2$ . By the inductive hypothesis,

$$\bar{v}(\psi_1) = \bar{\mu}(\psi_1) \quad \text{and} \quad \bar{v}(\psi_2) = \bar{\mu}(\psi_2),$$

so  $\bar{v}(\varphi) = \bar{\mu}(\varphi)$ .

### 7.3 Expressive Completeness Theorem

**Thm (Expressive Completeness).** For every  $n$  and every  $n$ -ary Boolean function  $f$ , there is some formula  $\varphi$  such that

$$f = f_\varphi^n.$$

**Motivation.** A Boolean function is a very general notion of a way to combine several statements into a new one in a way that only depends on the truth values of those statements, not their content. This theorem says that anything that can be thought of as a Boolean function can already be expressed in our language.

**Comment.** For any formulas  $\varphi$  and  $\psi$ ,

$$\varphi \equiv \psi \iff f_\varphi^n = f_\psi^n.$$

So the semantic content of a formula is captured by the corresponding Boolean function.

#### Proof idea (example)

Consider the Boolean function  $f : \{T, F\}^2 \rightarrow \{T, F\}$  given by:

$$f(X, Y) = \begin{cases} T & \text{if } X \neq Y, \\ F & \text{else.} \end{cases}$$

Truth table:

| $X$ | $Y$ | $f(X, Y)$ |
|-----|-----|-----------|
| T   | T   | F         |
| T   | F   | T         |
| F   | T   | T         |
| F   | F   | F         |

$f(X, Y)$  is true iff  $(X, Y)$  matches one of the rows where the output is T, namely:

$$(X = T, Y = F) \quad \text{or} \quad (X = F, Y = T).$$

Translate these into formulas:

$$(X = T, Y = F) \rightsquigarrow A_0 \wedge (\neg A_1), \quad (X = F, Y = T) \rightsquigarrow (\neg A_0) \wedge A_1.$$

Let

$$\varphi := (A_0 \wedge \neg A_1) \vee ((\neg A_0) \wedge A_1).$$

Then  $f_\varphi^2 = f$ .

## 7.4 The Proof

Suppose  $f : \{T, F\}^n \rightarrow \{T, F\}$  is an  $n$ -ary Boolean function.

For each  $\bar{X} = (X_0, \dots, X_{n-1}) \in \{T, F\}^n$  and each  $i < n$ , define

$$\theta_{\bar{X},i} := \begin{cases} A_i & \text{if } X_i = T, \\ \neg A_i & \text{if } X_i = F. \end{cases}$$

$$\theta_{\bar{X}} := \bigwedge_{i < n} \theta_{\bar{X},i}.$$

Example: if  $\bar{X} = (T, F, T, T)$ , then

$$\theta_{\bar{X}} = A_0 \wedge (\neg A_1) \wedge A_2 \wedge A_3.$$

Define

$$A = \{\bar{X} \in \{T, F\}^n \mid f(\bar{X}) = T\}$$

$$\varphi = \bigvee_{\bar{X} \in A} \theta_{\bar{X}}.$$

Example: if  $A = \{(T, F, T, T), (T, T, F, F)\}$ , then

$$\varphi = (A_0 \wedge \neg A_1 \wedge A_2 \wedge A_3) \vee (A_0 \wedge A_1 \wedge \neg A_2 \wedge \neg A_3).$$

**Claim:**  $f = f_{\varphi}^n$  (i.e. for all  $\bar{X}$ ,  $f(\bar{X}) = f_{\varphi}^n(\bar{X})$ .)

**Subclaim 1.** For all  $\bar{X} \in \{T, F\}^n$ ,  $i < n$ , and truth assignments  $v$ ,

$$\bar{v}(\theta_{\bar{X},i}) = T \iff v(A_i) = X_i.$$

**pf.** If  $X_i = T$  then  $\theta_{\bar{X},i} = A_i$ , so  $\bar{v}(\theta_{\bar{X},i}) = T \iff v(A_i) = T = X_i$

If  $X_i = F$  then  $\theta_{\bar{X},i} = \neg A_i$ , so  $\bar{v}(\theta_{\bar{X},i}) = T \iff v(A_i) = F = X_i$

**Subclaim 2.** For all  $\bar{X} \in \{T, F\}^n$  and truth assignments  $v$ ,

$$\bar{v}(\theta_{\bar{X}}) = T \iff \forall i < n, v(A_i) = X_i.$$

**pf.** Since  $\theta_{\bar{X}} = \theta_{\bar{X},0} \wedge \cdots \wedge \theta_{\bar{X},n-1}$ ,

$$\bar{v}(\theta_{\bar{X}}) = \text{T} \iff \forall i < n, \bar{v}(\theta_{\bar{X},i}) = \text{T} \iff \forall i < n, v(A_i) = X_i$$

by Subclaim 1.

**Proof of Claim** Fix  $\bar{X} \in \{\text{T}, \text{F}\}^n$  and let  $v$  be any truth assignment such that  $\forall i < n, v(A_i) = X_i$

Goal: Show  $f(\bar{X}) = \bar{v}(\varphi)$ .

Since  $\varphi = \bigvee_{\bar{Y} \in A} \theta_{\bar{Y}}$ ,

$$\bar{v}(\varphi) = \text{T} \iff \exists \bar{Y} \in A, \bar{v}(\theta_{\bar{Y}}) = \text{T} \stackrel{\text{Subcl. 2}}{\iff} \exists \bar{Y} \in A, \forall i < n, v(A_i) = Y_i.$$

By definition of  $v(\forall i < n, v(A_i) = X_i)$ , so the last condition is equivalent to  $\exists \bar{Y} \in A$  with  $\bar{X} = \bar{Y}$ , i.e.  $\bar{X} \in A$ . By definition of  $A$ ,

$$\bar{X} \in A \iff f(\bar{X}) = \text{T}.$$

So  $\bar{v}(\varphi) = \text{T} \iff f(\bar{X}) = \text{T}$ , i.e.  $\bar{v}(\varphi) = f(\bar{X})$ . Thus  $f = f_\varphi^n$

## 7.5 Comments about the proof

1. Given  $f : \{\text{T}, \text{F}\}^n \rightarrow \{\text{T}, \text{F}\}$ , the formula  $\varphi$  produced by our proof has size up to

$$(\text{number of tuples}) \times (\text{approx. length of each } \theta_{\bar{X}}) \approx 2^n \times n.$$

This is very large (for  $n = 10$ ,  $\approx 10,000$ ).

2. The formula we found had the form

$$(\theta_{0,0} \wedge \theta_{0,1} \wedge \cdots \wedge \theta_{0,n-1}) \vee \cdots \vee (\theta_{m,0} \wedge \theta_{m,1} \wedge \cdots \wedge \theta_{m,n-1}),$$

where each  $\theta_{i,j}$  is either a propositional symbol or the negation of a propositional symbol. This is called **disjunctive normal form**. Also, this shows that every formula is logically equivalent to one of this form.

## 8. Proofs

### 8.1 Proofs in propositional logic

**Question:** Suppose you know that some formula  $\varphi$  is a tautology. How could you convince someone else?

**Answer 1:** Write down the truth table of  $\varphi$ .

**Problem:** The truth table can be enormous! Size up to  $\approx 2^{101}$ .

E.g. If

$$\varphi = A_0 \vee (\neg A_0) \vee A_1 \vee A_2 \vee A_3 \vee A_4 \vee A_5 \vee A_6 \vee A_7,$$

then it's easy to explain why  $\varphi$  is always true, but the truth table has 256 rows.

**Question:** Is there another method which is shorter at least sometimes (e.g. when there is a simple reason why  $\varphi$  is always true)?

**Answer:** Proof system for propositional logic (next 2 lectures).

**Goals for a proof system:**

1. There should be a simple and efficient algorithm to check that a proof is valid.
2. (**Soundness**) If there is a proof of  $\varphi$  then  $\varphi$  really is a tautology.
3. (**Completeness**) If  $\varphi$  is a tautology then there is a proof of  $\varphi$ .
4. Proofs should be easy to construct — if you really understand why  $\varphi$  is true it should be easy to write a proof of  $\varphi$ .

**So,** Proofs should resemble informal mathematical proofs as much as possible.

### 8.2 A proof system

**Question:** What is an (informal) mathematical proof?

**Answer:**

1. A proof is a list of assertions (often implicit / not spelled out).

2. Each assertion claims that some conclusion follows from some assumptions.
3. Each assertion is justified using the previous assertions in the proof.

**Example:** If  $n$  is an integer then  $n(n + 1)$  is even.

**Proof:** First suppose  $n$  is even. Then  $n = 2k$  for some  $k$ . So

$$n(n + 1) = 2k(2k + 1) = 2(k(2k + 1)),$$

so  $n(n + 1)$  is even by definition.

Now suppose  $n$  is odd. Then  $n = 2k + 1$  for some  $k$ . So

$$n(n + 1) = (2k + 1)(2k + 2) = 2((2k + 1)(k + 1)),$$

so  $n(n + 1)$  is even.

So in either case,  $n(n + 1)$  is even.

We will describe a formal (i.e. mathematically precise) version of this.

The formal version of “**assertions**” (including the implicit assumptions) will be called “**sequents**”.

The formal version of “**justifications**” will be called “**rules of inference**”.

### 8.2.1 Sequents

**Definition:** A **sequent** is a pair  $\langle \Gamma, \varphi \rangle$  consisting of a set of formulas  $\Gamma$  and a formula  $\varphi$ .

We often write a sequent  $\langle \Gamma, \varphi \rangle$  as

$$\Gamma \vdash \varphi.$$

**Examples:**

$$\emptyset \vdash A \vee \neg A, \quad \{A, B\} \vdash A \wedge B, \quad \{A, B, C, D, E\} \vdash A \vee B.$$

**Idea:**

$$\Gamma \vdash \varphi$$



should be thought of as the assertion that if every formula in  $\Gamma$  is true then  $\varphi$  is true too.

i.e.  $\Gamma \vdash \varphi$  asserts that  $\Gamma \models \varphi$ .

### 8.2.2 Inference rules

Each inference rule states that if one or more sequents hold then some other sequent also holds.

**Examples (convenient notation):**

$$\frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \varphi} \quad (\text{usually called “and elimination”})$$

This should be interpreted as saying that for any set of formulas  $\Gamma$  and any formulas  $\varphi, \psi$ , if we know  $\Gamma \vdash \varphi \wedge \psi$  then we can conclude that  $\Gamma \vdash \varphi$ .

$$\frac{\Gamma \vdash \varphi \quad \Gamma \vdash \psi}{\Gamma \vdash \varphi \wedge \psi} \quad (\text{usually called “and introduction”})$$

This should be interpreted as saying that for any set of formulas  $\Gamma$  and any formulas  $\varphi, \psi$ , if we know  $\Gamma \vdash \varphi$  and we know  $\Gamma \vdash \psi$  then we can conclude that  $\Gamma \vdash \varphi \wedge \psi$ .

Most of the logical symbols have two corresponding rules of inference:

- **Introduction rule:** tells you how to prove a formula involving that symbol.
- **Elimination rule:** tells you how to use a formula involving that symbol.

**Example ( $\rightarrow$ ):**

$\rightarrow$ -**Introduction:** to prove  $\varphi \rightarrow \psi$ , assume  $\varphi$  and prove  $\psi$ :

$$\frac{\Gamma \cup \{\varphi\} \vdash \psi}{\Gamma \vdash \varphi \rightarrow \psi}.$$

$\rightarrow$ -**Elimination (modus ponens):**

$$\frac{\Gamma \vdash \varphi \rightarrow \psi \quad \Gamma \vdash \varphi}{\Gamma \vdash \psi}.$$

### 8.2.3 A complete list of inference rules

|                                  |                                                                                                                                  |                                                                                                                                                     |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Axiom:</b>                    | $\Gamma \vdash \varphi$ where $\varphi \in \Gamma$                                                                               | <b>Contradiction:</b> $\frac{\Gamma \cup \{\neg\varphi\} \vdash \perp}{\Gamma \vdash \varphi}$                                                      |
| <b>Introduction</b>              |                                                                                                                                  | <b>Elimination</b>                                                                                                                                  |
| <b>And</b> ( $\wedge$ )          | $\frac{\Gamma \vdash \varphi \quad \Gamma \vdash \psi}{\Gamma \vdash \varphi \wedge \psi}$                                       | $\frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \varphi} \quad \frac{\Gamma \vdash \varphi \wedge \psi}{\Gamma \vdash \psi}$                |
| <b>Or</b> ( $\vee$ )             | $\frac{\Gamma \vdash \varphi}{\Gamma \vdash \varphi \vee \psi} \quad \frac{\Gamma \vdash \psi}{\Gamma \vdash \varphi \vee \psi}$ | $\frac{\Gamma \vdash \varphi \vee \psi \quad \Gamma \cup \{\varphi\} \vdash \theta \quad \Gamma \cup \{\psi\} \vdash \theta}{\Gamma \vdash \theta}$ |
| <b>Not</b> ( $\neg$ )            | $\frac{\Gamma \cup \{\varphi\} \vdash \perp}{\Gamma \vdash \neg\varphi}$                                                         | $\frac{\Gamma \vdash \neg\varphi \quad \Gamma \vdash \varphi}{\Gamma \vdash \perp}$                                                                 |
| <b>Implies</b> ( $\rightarrow$ ) | $\frac{\Gamma \cup \{\varphi\} \vdash \psi}{\Gamma \vdash \varphi \rightarrow \psi}$                                             | $\frac{\Gamma \vdash \varphi \rightarrow \psi \quad \Gamma \vdash \varphi}{\Gamma \vdash \psi}$                                                     |
| $\top/\perp$                     | $\Gamma \vdash \top$                                                                                                             | $\frac{\Gamma \vdash \perp}{\Gamma \vdash \varphi}$                                                                                                 |

The following are some notable inference rules from the list above that are commonly referred to by specific names:

- **Contradiction:** Known as *proof by contradiction*, it allows deriving  $\varphi$  if assuming  $\neg\varphi$  leads to a contradiction ( $\perp$ ).
- **$\vee$ -Elimination:** Known as *proof by cases*, it proves  $\theta$  by showing it follows from both possible cases,  $\varphi$  and  $\psi$ .
- **$\rightarrow$ -Elimination:** Known as *Modus Ponens*, it derives  $\psi$  from the conditional  $\varphi \rightarrow \psi$  and the antecedent  $\varphi$ .
- **$\perp$ -Elimination:** Known as the *explosion principle*, it states that from a contradiction ( $\perp$ ), any arbitrary formula  $\varphi$  can be derived.

### 8.3 Proofs

**Definition:** A **proof** is a list of sequents

$$\langle \Gamma_0 \vdash \varphi_0, \Gamma_1 \vdash \varphi_1, \dots, \Gamma_n \vdash \varphi_n \rangle$$

such that each sequent can be justified using an inference rule together with sequents earlier in the list.

Formally, for each  $i \leq n$ , one of the following holds (these correspond to the 14 inference rules):

1.  $\varphi_i \in \Gamma_i$  (*i.e.*  $\Gamma_i \vdash \varphi_i$  is justified by the axiom rule)
2. For some  $j < i$ ,  $\Gamma_j = \Gamma_i \cup \{\neg\varphi_i\}$  and  $\varphi_j = \perp$
3. For some  $j, k < i$ ,  $\Gamma_j = \Gamma_k = \Gamma_i$  and  $\varphi_i = \varphi_j \wedge \varphi_k$
- $\vdots$
14. For some  $j < i$ ,  $\Gamma_j = \Gamma_i$  and  $\varphi_j = \perp$  (*i.e.*  $\Gamma_i \vdash \varphi_i$  can be justified using  $\perp$ -elimination together with  $\Gamma_j = \perp$  (which is  $\Gamma_i \vdash \perp$ ))

**Definition:** A set of formulas,  $\Gamma$ , **proves** a formula  $\varphi$ , written  $\Gamma \vdash \varphi$ , if there is a proof

$$\langle \Gamma_0 \vdash \varphi_0, \dots, \Gamma_n \vdash \varphi_n \rangle$$

such that  $\Gamma = \Gamma_n$  and  $\varphi = \varphi_n$ .

**Confusing point:**  $\Gamma \vdash \varphi$  has two different meanings:

1.  $\Gamma \vdash \varphi$  can be a line in a proof.
2.  $\Gamma \vdash \varphi$  can express the fact that  $\Gamma$  proves  $\varphi$  (the previous definition).

## 8.4 Examples

(1)  $\{(A \wedge (B \wedge C))\} \vdash C$

Proof:

- |    |                                                            |                       |
|----|------------------------------------------------------------|-----------------------|
| 0. | $\{(A \wedge (B \wedge C))\} \vdash A \wedge (B \wedge C)$ | Axiom                 |
| 1. | $\{(A \wedge (B \wedge C))\} \vdash (B \wedge C)$          | $\wedge$ -Elimination |
| 2. | $\{(A \wedge (B \wedge C))\} \vdash C$                     | $\wedge$ -Elimination |

(2)  $\{\neg(A \wedge B), A\} \vdash \neg B$

Proof:

- |    |                                                      |                        |
|----|------------------------------------------------------|------------------------|
| 0. | $\{\neg(A \wedge B), A, B\} \vdash A$                | Axiom                  |
| 1. | $\{\neg(A \wedge B), A, B\} \vdash B$                | Axiom                  |
| 2. | $\{\neg(A \wedge B), A, B\} \vdash (A \wedge B)$     | $\wedge$ -Introduction |
| 3. | $\{\neg(A \wedge B), A, B\} \vdash \neg(A \wedge B)$ | Axiom                  |
| 4. | $\{\neg(A \wedge B), A, B\} \vdash \perp$            | $\neg$ -Elimination    |
| 5. | $\{\neg(A \wedge B), A\} \vdash \neg B$              | $\neg$ -Introduction   |

(3)  $\{A \vee C, B \vee \neg C\} \vdash A \vee B$ .

Proof:

0. Try yourself!

## 8.5 Soundness and Completeness

In later lectures we will prove :

For any  $\Gamma$  and  $\varphi$ :

**Thm (Soundness):** If  $\Gamma \vdash \varphi$  then  $\Gamma \models \varphi$ .

**Thm (Completeness):** If  $\Gamma \models \varphi$  then  $\Gamma \vdash \varphi$ .

In other words,  $\Gamma \models \varphi$  if and only if there is a proof of  $\varphi$  from  $\Gamma$ .

## 9. Soundness

### 9.1 Introduction

We now have several ways to say that  $\varphi$  implies  $\psi$ :

1.  $\varphi \rightarrow \psi$  is a tautology. (For all  $v$ ,  $\bar{v}(\varphi \rightarrow \psi) = T$ )
2.  $\{\varphi\} \models \psi$ . (For all  $v$ , if  $\bar{v}(\varphi) = T$  then  $\bar{v}(\psi) = T$ )
3.  $\{\varphi\} \vdash \psi$ . (There is a proof of  $\psi$  from  $\varphi$ )

Actually, these are all equivalent.

- $\textcircled{1} \Leftrightarrow \textcircled{2}$  follows from the definition of  $\bar{v}(\varphi \rightarrow \psi)$ .
- $\textcircled{2} \Rightarrow \textcircled{3}$ : **Completeness thm**
- $\textcircled{3} \Rightarrow \textcircled{2}$ : **Soundness thm**

$\textcircled{2} \Leftrightarrow \textcircled{3}$  More generally,  $\Gamma \models \varphi \Leftrightarrow \Gamma \vdash \varphi$ .

### 9.2 Soundness and Completeness

In the next 2–3 lectures, we will prove:

**Thm (Soundness):** For all sets of formulas,  $\Gamma$ , and formulas  $\varphi$ , if  $\Gamma \vdash \varphi$  then  $\Gamma \models \varphi$ .

*“You can’t prove false things”*

**Thm (Completeness):** For all sets of formulas,  $\Gamma$ , and formulas  $\varphi$ , if  $\Gamma \models \varphi$  then  $\Gamma \vdash \varphi$ .

*“Everything true is provable”*

**Cor:**  $\varphi$  is a tautology if and only if  $\emptyset \vdash \varphi$ .

- **Soundness:** Proof is a straightforward induction with many cases.
- **Completeness:** Proof requires some new ideas.

### 9.3 Applications of Soundness & Completeness

Sometimes it's easy to prove  $\Gamma \models \varphi$  ( $\Gamma \not\models \varphi$ ) and hard to prove  $\Gamma \vdash \varphi$  ( $\Gamma \not\vdash \varphi$ ).

Sometimes it's easy to prove  $\Gamma \vdash \varphi$  ( $\Gamma \not\vdash \varphi$ ) and hard to prove  $\Gamma \models \varphi$  ( $\Gamma \not\models \varphi$ ).

Soundness and completeness let us switch back-and-forth and use whichever of  $\Gamma \models \varphi$ ,  $\Gamma \vdash \varphi$  is easier.

**Example:** Show that  $\{A_0 \vee A_1\} \not\models A_0 \wedge A_1$ .

We could try to analyze all proofs...

But this is easier: Define

$$v(A_i) = \begin{cases} T & \text{if } i = 0 \\ F & \text{if } i > 0 \end{cases}$$

$$\begin{aligned} \bar{v}(A_0 \vee A_1) = T \\ \bar{v}(A_0 \wedge A_1) = F \end{aligned} \Rightarrow \{A_0 \vee A_1\} \not\models A_0 \wedge A_1 \Rightarrow \{A_0 \vee A_1\} \not\models A_0 \wedge A_1$$

### 9.4 Proof of Soundness

**Thm (Soundness):** For all sets of formulas,  $\Gamma$ , and formulas  $\varphi$ , if  $\Gamma \vdash \varphi$  then  $\Gamma \models \varphi$ .

**Proof:** Suppose  $\langle \Gamma_0 \vdash \varphi_0, \Gamma_1 \vdash \varphi_1, \dots, \Gamma_n \vdash \varphi_n \rangle$  is a proof of  $\Gamma \vdash \varphi$ .

**Key idea:** Show by induction on  $i$  that  $\Gamma_i \models \varphi_i$ .

**Base case:**  $\Gamma_0 \vdash \varphi_0$  must either be an application of Axiom or  $\top$ -introduction.

- **Axiom:**  $\varphi_0 \in \Gamma_0 \Rightarrow \Gamma_0 \models \varphi_0$ .
- **$\top$ -introduction:**  $\varphi_0 = \top \Rightarrow \Gamma_0 \models \varphi_0$ .

**Inductive case:** Assume  $\forall j < i, \Gamma_j \models \varphi_j$ .

Want to show:  $\Gamma_i \models \varphi_i$ .

14 cases to check, one for each rule of inference.

## 9.5 A few representative cases

**$\wedge$ -Introduction:**

$$\frac{\Gamma \vdash \varphi \quad \Gamma \vdash \psi}{\Gamma \vdash \varphi \wedge \psi}$$

Need to check: If  $\Gamma \models \varphi$  and  $\Gamma \models \psi$  then  $\Gamma \models \varphi \wedge \psi$ .

**pf:** Suppose  $\Gamma \models \varphi$ ,  $\Gamma \models \psi$  and let  $v$  be a truth assignment such that  $v \models \Gamma$ .

$$\Gamma \models \varphi, \Gamma \models \psi \Rightarrow \bar{v}(\varphi) = \bar{v}(\psi) = T \Rightarrow \bar{v}(\varphi \wedge \psi) = T.$$

**$\vee$ -Elimination:**

$$\frac{\Gamma \vdash \varphi \vee \psi \quad \Gamma \cup \{\varphi\} \vdash \theta \quad \Gamma \cup \{\psi\} \vdash \theta}{\Gamma \vdash \theta}$$

Need to check: If  $\Gamma \models \varphi \vee \psi$ ,  $\Gamma \cup \{\varphi\} \models \theta$  and  $\Gamma \cup \{\psi\} \models \theta$  then  $\Gamma \models \theta$ .

**pf:** Suppose  $v \models \Gamma$ .

$$\Gamma \models \varphi \vee \psi \Rightarrow \bar{v}(\varphi \vee \psi) = T \Rightarrow \bar{v}(\varphi) = T \text{ or } \bar{v}(\psi) = T.$$

- **Case 1:**  $\bar{v}(\varphi) = T \Rightarrow v \models \Gamma \cup \{\varphi\} \Rightarrow \bar{v}(\theta) = T$ .
- **Case 2:**  $\bar{v}(\psi) = T \Rightarrow v \models \Gamma \cup \{\psi\} \Rightarrow \bar{v}(\theta) = T$ .

## 10. Completeness

### 10.1 Consistency and satisfiability

**Goal:** Prove that our proof system is complete, i.e.

$$\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi.$$

**Def:** A set of formulas,  $\Gamma$ , is:

- **inconsistent** if  $\Gamma \vdash \perp$ ,
- **consistent** if  $\Gamma \not\vdash \perp$ ,
- **satisfiable** if for some  $v$ ,  $v \models \Gamma$  (i.e. for all  $\psi \in \Gamma$ ,  $\bar{v}(\psi) = T$ ),
- **unsatisfiable** if for every  $v$ ,  $v \not\models \Gamma$ .

**Example:**  $\Gamma = \{A_0 \vee A_1, \neg A_0, \neg A_1\}$  is unsatisfiable.

If  $v \models \Gamma$  then  $\bar{v}(A_0) = \bar{v}(A_1) = F \Rightarrow \bar{v}(A_0 \vee A_1) = F$ .

inconsistent:

0.  $\Gamma \cup \{A_0\} \vdash A_0$
1.  $\Gamma \cup \{A_0\} \vdash \neg A_0$
2.  $\Gamma \cup \{A_0\} \vdash \perp$  ( $\neg$ -Elim)
3.  $\Gamma \cup \{A_1\} \vdash A_1$
4.  $\Gamma \cup \{A_1\} \vdash \neg A_1$
5.  $\Gamma \cup \{A_1\} \vdash \perp$  ( $\neg$ -Elim)
6.  $\Gamma \vdash A_0 \vee A_1$
7.  $\Gamma \vdash \perp$  ( $\vee$ -Elim, using 2 and 5)



## 10.2 How to prove completeness

Instead of proving completeness directly, we will prove:

**Thm 10.1 (Completeness, version 2):** For any set of formulas,  $\Gamma$ , if  $\Gamma$  is consistent then  $\Gamma$  is satisfiable.

The following two lemmas imply that completeness follows from Completeness v.2:

**Lemma 10.1:** If  $\Gamma \not\vdash \varphi$  then  $\Gamma \cup \{\neg\varphi\}$  is consistent.

**Lemma 10.2:** If  $\Gamma \cup \{\neg\varphi\}$  is satisfiable then  $\Gamma \not\models \varphi$ .

**Proof of completeness from completeness v.2:**

Completeness:

$$\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi.$$

Contrapositive:

$$\Gamma \not\vdash \varphi \Rightarrow \Gamma \not\models \varphi.$$

Assume  $\Gamma \not\vdash \varphi$ .

By the Lemma 10.1,  $\Gamma \cup \{\neg\varphi\}$  is consistent.

By completeness v.2,  $\Gamma \cup \{\neg\varphi\}$  is satisfiable.

By the Lemma 10.2,  $\Gamma \not\models \varphi$ .

**Lemma 10.1:** If  $\Gamma \not\vdash \varphi$  then  $\Gamma \cup \{\neg\varphi\}$  is consistent.

**Proof (contrapositive):**  $\Gamma \cup \{\neg\varphi\}$  is inconsistent  $\Rightarrow \Gamma \vdash \varphi$

Assume  $\Gamma \cup \{\neg\varphi\}$  is inconsistent.

By def.,  $\Gamma \cup \{\neg\varphi\} \vdash \perp$ .

Let  $\langle \Gamma_0 \vdash \varphi_0, \dots, \Gamma_n \vdash \varphi_n \rangle$  be a proof of  $\Gamma \cup \{\neg\varphi\} \vdash \perp$ .

Then using the contradiction rule,  $\langle \Gamma_0 \vdash \varphi_0, \dots, \Gamma_n \vdash \varphi_n, \Gamma \vdash \varphi \rangle$  is a proof of  $\Gamma \vdash \varphi$ .

(Proof sketch as a list:)

0.  $\Gamma_0 \vdash \varphi_0$

$\vdots$

$n.$   $\Gamma \cup \{\neg\varphi\} \vdash \perp$   
 $n+1.$   $\Gamma \vdash \varphi$  (Contradiction rule from line  $n$ )

**Lemma 10.2:** If  $\Gamma \cup \{\neg\varphi\}$  is satisfiable then  $\Gamma \not\models \varphi$ .

**Proof:**  $\Gamma \cup \{\neg\varphi\}$  satisfiable  $\Rightarrow$  there is some  $v$  such that  $v \models \Gamma \cup \{\neg\varphi\}$ .

$v \models \Gamma \cup \{\neg\varphi\} \Rightarrow v \models \Gamma$

$v \models \Gamma \cup \{\neg\varphi\} \Rightarrow \bar{v}(\neg\varphi) = T \Rightarrow \bar{v}(\varphi) = F$ .

So  $v \models \Gamma$  but  $\bar{v}(\varphi) \neq T$ , i.e.  $\Gamma \not\models \varphi$ .

### 10.3 Proof strategy

Original goal:

$$\Gamma \models \varphi \Rightarrow \Gamma \vdash \varphi.$$

New goal:

$$\Gamma \text{ consistent} \Rightarrow \Gamma \text{ satisfiable.}$$

**Key definition:** A set of formulas  $\Gamma$  is **maximally consistent** if:

1.  $\Gamma$  is consistent
2. and for all formulas  $\varphi$ , either  $\varphi \in \Gamma$  or  $\neg\varphi \in \Gamma$ .

**Examples:**

- $\Gamma_0 =$  the set of all formulas. Not consistent (since  $\perp \in \Gamma_0$  so  $\Gamma_0 \vdash \perp$ ).
- $\Gamma_1 = \{A_0 \vee A_1\}$ . Consistent, but not maximally consistent (neither  $A_0$  nor  $\neg A_0$  is in  $\Gamma_1$ ).
- $\Gamma_2 = \{\varphi \mid \bar{v}(\varphi) = T\}$  for some fixed  $v$ . This is maximally consistent: Soundness  $\Rightarrow$  Consistent. For every  $\varphi$ , either  $\bar{v}(\varphi) = T$  or  $\bar{v}(\neg\varphi) = T$ ,  $\Rightarrow$  max.consistent

**Idea of proof that  $\Gamma$  consistent  $\Rightarrow \Gamma$  satisfiable:**

1. Start with  $\Gamma$  consistent.
2. Find  $\Gamma' \supseteq \Gamma$  such that  $\Gamma'$  is maximally consistent.

3. Show any maximally consistent set is satisfiable,  $\rightsquigarrow$  get  $v$  such that  $v \models \Gamma'$ .
4. Since  $\Gamma \subseteq \Gamma'$ ,  $v \models \Gamma'$  we get  $v \models \Gamma$ .

Hence  $\Gamma$  is satisfiable.

Formally, the theorem follows from the following two thms:

- **Thm 10.2:** If  $\Gamma$  is consistent then there is some maximally consistent set  $\Gamma'$  with  $\Gamma \subseteq \Gamma'$ .
- **Thm 10.3:** If  $\Gamma$  is maximally consistent then it is satisfiable.

We will prove the theorem 10.3 first.

## 10.4 Satisfiability from maximal consistency

**Thm 10.3:** If  $\Gamma$  is maximally consistent then  $\Gamma$  is satisfiable.

To prove this, we need to build a truth assignment  $v$  such that for all  $\varphi \in \Gamma$ ,  $\bar{v}(\varphi) = T$

**Key idea:** Let  $\Gamma$  tell us how to define  $v$

**Proof:** Define a truth assignment  $v : \{A_0, A_1, \dots\} \rightarrow \{T, F\}$  by

$$v(A_i) = \begin{cases} T & \text{if } A_i \in \Gamma, \\ F & \text{else} \end{cases}$$

We will show by structural induction that for all formulas  $\varphi$ ,

$$\bar{v}(\varphi) = T \iff \varphi \in \Gamma.$$

This implies  $v \models \Gamma$ .

The following lemma will be useful in the proof.

**Lemma 10.3:** If  $\Gamma$  is maximally consistent and  $\Gamma \vdash \varphi$ , then  $\varphi \in \Gamma$ .

**Proof:** Suppose for contradiction that  $\Gamma \vdash \varphi$  but  $\varphi \notin \Gamma$ .

Since  $\Gamma$  is maximally consistent,  $\Rightarrow \neg\varphi \in \Gamma$ .

Let  $\langle \Gamma_0 \vdash \varphi_0, \dots, \Gamma_n \vdash \varphi_n, \Gamma \vdash \varphi \rangle$  be a proof of  $\Gamma \vdash \varphi$

Then  $\langle \Gamma_0 \vdash \varphi_0, \dots, \Gamma_n \vdash \varphi_n, \Gamma \vdash \varphi, \Gamma \vdash \neg\varphi, \Gamma \vdash \perp \rangle$  is a proof of  $\Gamma \vdash \perp$

So  $\Gamma$  is not consistent.

(Proof sketch as a list:)

- $n.$   $\Gamma \vdash \varphi$  (given)
- $n+1.$   $\Gamma \vdash \neg\varphi$  (axiom since  $\neg\varphi \in \Gamma$ )
- $n+2.$   $\Gamma \vdash \perp$  ( $\neg$ -Elim from  $n, n+1$ )

**pf of thm (continued)**  $\Gamma =$  maximally consistent set

$$v(A_i) = T \iff A_i \in \Gamma$$

**Claim:** For all  $\varphi$ ,  $\bar{v}(\varphi) = T \iff \varphi \in \Gamma$ .

**pf:** By structural induction on  $\varphi$ .

**Base Cases:**

- ①  $\varphi = A_i$ :  $\bar{v}(A_i) = T \iff v(A_i) = T \iff A_i \in \Gamma$  (by def. of  $v$ )
- ②  $\varphi = \top$ :  $\bar{v}(\top) = T$  and  $\top \in \Gamma$  (since  $\Gamma \vdash \top$  by  $\top$ -introduction rule).
- ③  $\varphi = \perp$ :  $\bar{v}(\perp) = F$  and  $\perp \notin \Gamma$  (since otherwise  $\Gamma \vdash \perp$ , so  $\Gamma$  is not consistent).

**Inductive cases:** Assume  $\varphi, \psi$  satisfy  $\bar{v}(\varphi) = T \iff \varphi \in \Gamma$  and  $\bar{v}(\psi) = T \iff \psi \in \Gamma$ .

Need to show:

- 1.  $\bar{v}(\varphi \wedge \psi) = T \iff \varphi \wedge \psi \in \Gamma$
- 2.  $\bar{v}(\varphi \vee \psi) = T \iff \varphi \vee \psi \in \Gamma$  (Next 4 slides)
- 3.  $\bar{v}(\varphi \rightarrow \psi) = T \iff \varphi \rightarrow \psi \in \Gamma$
- 4.  $\bar{v}(\neg\varphi) = T \iff \neg\varphi \in \Gamma$

### 10.4.1 And case

**Assume:**  $\Gamma$  max. consistent,  $\bar{v}(\varphi) = T \Leftrightarrow \varphi \in \Gamma$  and  $\bar{v}(\psi) = T \Leftrightarrow \psi \in \Gamma$  (IH)

**WTS:**  $\bar{v}(\varphi \wedge \psi) = T \Leftrightarrow \varphi \wedge \psi \in \Gamma$ .

( $\Rightarrow$ ) Assume  $\bar{v}(\varphi \wedge \psi) = T$ .

$$\begin{aligned} &\Rightarrow \bar{v}(\varphi) = T \text{ and } \bar{v}(\psi) = T \\ &\Rightarrow \varphi \in \Gamma \text{ and } \psi \in \Gamma \quad (\text{By IH}) \\ &\Rightarrow \Gamma \vdash \varphi \wedge \psi \quad (\wedge\text{-Intro}) \\ &\Rightarrow \varphi \wedge \psi \in \Gamma \quad (\text{By lemma}) \end{aligned}$$

*proof:*

- |                                        |                 |
|----------------------------------------|-----------------|
| 0. $\Gamma \vdash \varphi$             | Axiom           |
| 1. $\Gamma \vdash \psi$                | Axiom           |
| 2. $\Gamma \vdash \varphi \wedge \psi$ | $\wedge$ -Intro |

( $\Leftarrow$ ) Assume  $\varphi \wedge \psi \in \Gamma$ .

$$\begin{aligned} &\Rightarrow \Gamma \vdash \varphi \text{ and } \Gamma \vdash \psi \\ &\Rightarrow \varphi \in \Gamma \text{ and } \psi \in \Gamma \quad (\text{By lemma}) \\ &\Rightarrow \bar{v}(\varphi) = T \text{ and } \bar{v}(\psi) = T \quad (\text{By IH}) \\ &\Rightarrow \bar{v}(\varphi \wedge \psi) = T \end{aligned}$$

*proof:*

- |                                        |                 |
|----------------------------------------|-----------------|
| 0. $\Gamma \vdash \varphi \wedge \psi$ | Axiom           |
| 1. $\Gamma \vdash \varphi$             | $\wedge$ -Elim. |

### 10.4.2 Or case

**WTS:**  $\bar{v}(\varphi \vee \psi) = T \Leftrightarrow \varphi \vee \psi \in \Gamma$ .

( $\Rightarrow$ ) Assume  $\bar{v}(\varphi \vee \psi) = T \Rightarrow \bar{v}(\varphi) = T$  or  $\bar{v}(\psi) = T$ .

$$\begin{aligned}
\bar{v}(\varphi) = T &\Rightarrow \varphi \in \Gamma \\
&\Rightarrow \Gamma \vdash \varphi \vee \psi \quad (\text{By } \vee\text{-Intro}) \\
&\Rightarrow \varphi \vee \psi \in \Gamma
\end{aligned}$$

Case of  $\bar{v}(\psi) = T$  (case  $\bar{v}(\psi) = T$  is similar):

( $\Leftarrow$ ) Assume  $\varphi \vee \psi \in \Gamma$ . By IH, enough to show  $\varphi \in \Gamma$  or  $\psi \in \Gamma$ .

For contradiction, assume  $\varphi \notin \Gamma$  and  $\psi \notin \Gamma$ .

$$\begin{aligned}
\Gamma \text{ max. consistent} &\Rightarrow \neg\varphi \in \Gamma \text{ and } \neg\psi \in \Gamma \\
&\Rightarrow \Gamma \vdash \perp \quad (\text{Contradicts consistency of } \Gamma)
\end{aligned}$$

*proof:*

- |                                                 |               |
|-------------------------------------------------|---------------|
| 0. $\Gamma \vdash \varphi \vee \psi$            |               |
| 1. $\Gamma \cup \{\varphi\} \vdash \varphi$     |               |
| 2. $\Gamma \cup \{\varphi\} \vdash \neg\varphi$ |               |
| 3. $\Gamma \cup \{\varphi\} \vdash \perp$       | $\neg$ -Elim. |
| 4. $\Gamma \cup \{\psi\} \vdash \psi$           |               |
| 5. $\Gamma \cup \{\psi\} \vdash \neg\psi$       |               |
| 6. $\Gamma \cup \{\psi\} \vdash \perp$          | $\neg$ -Elim. |
| 7. $\Gamma \vdash \perp$                        | $\vee$ -Elim. |

### 10.4.3 Implies and Not cases

**Implies:** Exercise.

**Not: WTS:**  $\bar{v}(\neg\varphi) = T \Leftrightarrow \neg\varphi \in \Gamma$ .

( $\Rightarrow$ ) Assume  $\bar{v}(\neg\varphi) = T$ .

$$\begin{aligned}
&\Rightarrow \bar{v}(\varphi) = F \\
&\Rightarrow \varphi \notin \Gamma \quad (\text{By IH}) \\
&\Rightarrow \neg\varphi \in \Gamma \quad (\text{Since } \Gamma \text{ max. consistent})
\end{aligned}$$

( $\Leftarrow$ ) Assume  $\neg\varphi \in \Gamma$ .

$$\begin{aligned} &\Rightarrow \varphi \notin \Gamma \quad (\text{If } \varphi, \neg\varphi \in \Gamma \text{ then } \Gamma \text{ is inconsistent}) \\ &\Rightarrow \bar{v}(\varphi) \neq T \\ &\Rightarrow \bar{v}(\varphi) = F \\ &\Rightarrow \bar{v}(\neg\varphi) = T \end{aligned}$$

because:

- |                                |               |
|--------------------------------|---------------|
| 0. $\Gamma \vdash \varphi$     | Axiom         |
| 1. $\Gamma \vdash \neg\varphi$ | Axiom         |
| 2. $\Gamma \vdash \perp$       | $\neg$ -Elim. |

## 10.5 Finding maximally consistent sets

**Thm 10.2:** If  $\Gamma$  is consistent then there is some maximally consistent set  $\Gamma'$  such that  $\Gamma \subseteq \Gamma'$ .

**Idea:** Go through all the formulas one-by-one, and for each formula  $\varphi$  either add  $\varphi$  or  $\neg\varphi$  to  $\Gamma$ .

**Useful fact:** Since the set of all strings over a ctbl set is countable, the set of all well-formed formulas is countable.

Using this, we can safely assume we have an enumeration  $\varphi_0, \varphi_1, \varphi_2, \dots$  of all wffs.

**pf:** Let  $\varphi_0, \varphi_1, \varphi_2, \dots$  be an enumeration of all wffs. Inductively define a sequence  $\Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \dots$  by

$$\begin{aligned} \Gamma_0 &= \Gamma \\ \Gamma_{i+1} &= \begin{cases} \Gamma_i \cup \{\varphi_i\} & \text{if } \Gamma_i \cup \{\varphi_i\} \text{ is consistent} \\ \Gamma_i \cup \{\neg\varphi_i\} & \text{else} \end{cases} \end{aligned}$$

Set

$$\Gamma' = \bigcup_{i \in \mathbb{N}} \Gamma_i.$$

**Need to show:**

1.  $\Gamma \subseteq \Gamma'$
  2. For all  $\varphi$ , either  $\varphi \in \Gamma'$  or  $\neg\varphi \in \Gamma'$
  3.  $\Gamma'$  is consistent
- ① & ② are clear from the construction. So we just need to prove ③.

$$\Gamma = \Gamma_0 \subseteq \Gamma_1 \subseteq \Gamma_2 \subseteq \dots \quad \Gamma' = \bigcup_{i \in \mathbb{N}} \Gamma_i$$

**Claim:** For each  $i$ ,  $\Gamma_i$  is consistent.

**Facts from HW 4:**

1. If  $\Delta \subseteq \Delta'$  and  $\Delta \vdash \varphi$  then  $\Delta' \vdash \varphi$ .
2. If  $\Delta \vdash \varphi$  then there is some finite  $\Delta' \subseteq \Delta$  s.t.  $\Delta' \vdash \varphi$ .

**proof**  $\Gamma'$  is consistent using claim + HW 4:

Assume for contradiction that  $\Gamma'$  is inconsistent, i.e.  $\Gamma' \vdash \perp$ .

Fact ② from HW 4  $\Rightarrow$  for some finite  $\Delta \subseteq \Gamma'$ ,  $\Delta \vdash \perp$ .

$\Delta$  finite &  $\Gamma_0 \subseteq \Gamma_1 \subseteq \dots \Rightarrow$  for some  $N$ ,  $\Delta \subseteq \Gamma_N$ .

Fact ① from HW 4  $\Rightarrow \Gamma_N \vdash \perp$ . But this contradicts the claim.

**pf of Claim:**

By induction on  $i$ .

**Base case:**  $i = 0$ .  $\Gamma_0 = \Gamma$  which is consistent by assumption.

**Inductive case:** Assume  $\Gamma_i$  consistent. (IH)

**Case 1:**  $\Gamma_i \cup \{\varphi_i\}$  is consistent. Then  $\Gamma_{i+1} = \Gamma_i \cup \{\varphi_i\}$  is consistent.

**Case 2:**  $\Gamma_i \cup \{\varphi_i\}$  is inconsistent  $\Rightarrow \Gamma_{i+1} = \Gamma_i \cup \{\neg\varphi_i\}$ .

Assume for contradiction  $\Gamma_{i+1}$  inconsistent.

$$\begin{aligned} &\Rightarrow \Gamma_i \cup \{\varphi_i\} \vdash \perp \text{ and } \Gamma_i \cup \{\neg\varphi_i\} \vdash \perp \\ &\Rightarrow \Gamma_i \vdash \perp \quad (\text{Lemma 10.4}) \text{ It contradicts IH} \end{aligned}$$



**Lemma 10.4:** If  $\Gamma \cup \{\varphi\} \vdash \perp$  and  $\Gamma \cup \{\neg\varphi\} \vdash \perp$  then  $\Gamma \vdash \perp$ .

**pf:**

|                                                         |                                                        |
|---------------------------------------------------------|--------------------------------------------------------|
| 0. $\vdots$                                             | (proof of $\Gamma \cup \{\varphi\} \vdash \perp$ )     |
| $n.$ $\Gamma \cup \{\varphi\} \vdash \perp$             |                                                        |
| $n + 1.$ $\vdots$                                       | (proof of $\Gamma \cup \{\neg\varphi\} \vdash \perp$ ) |
| $n + m + 1.$ $\Gamma \cup \{\neg\varphi\} \vdash \perp$ |                                                        |
| $n + m + 2.$ $\Gamma \vdash \neg\varphi$                | $\neg$ -introduction of line $n$                       |
| $n + m + 3.$ $\Gamma \vdash \neg\neg\varphi$            | $\neg$ -introduction of line $n + m + 1$               |
| $n + m + 4.$ $\Gamma \vdash \perp$                      | $\neg$ -elimination of lines $n + m + 2, n + m + 3$    |

## 11. Syntax of First Order Logic

### 11.1 Introduction

Previously in this note: We studied propositional logic, a mathematical model of a very restricted form of logical reasoning.

**Goal for the rest of the note:** Study a mathematical model of a larger portion of logical reasoning which is, in principle, sufficient for all known mathematical proofs. Known as **first order logic** (also predicate logic, quantificational logic).

**Two comments:**

1. Requires a lot more setup/complicated definitions than propositional logic. But the payoff is worth it!
2. Proofs of many basic facts are very similar to analogous facts for prop. logic but with more cases. We will skip these proofs.

### 11.2 Motivating examples

In first order logic, we can talk & reason about mathematical structures.

**Good source of examples and intuition:** Abstract algebra.

**E.g.** A group consists of:

- A set  $G$
- A binary function  $\cdot : G \times G \rightarrow G$
- A unary function  $^{-1} : G \rightarrow G$
- A constant  $e \in G$

satisfying the following axioms:

1.  $\forall x, y, z (x \cdot (y \cdot z) = (x \cdot y) \cdot z)$
2.  $\forall x (e \cdot x = x \text{ and } x \cdot e = x)$
3.  $\forall x (x \cdot x^{-1} = e \text{ and } x^{-1} \cdot x = e)$

**Examples:**

1.  $(\mathbb{Z}, +, -, 0)$
2.  $(\{f : \mathbb{N} \rightarrow \mathbb{N} \mid f \text{ is a bijection}\}, \circ, ^{-1}, \text{id}_{\mathbb{N}})$
3.  $(\{n \times n \text{ invertible matrices}\}, \cdot, ^{-1}, I_n)$

We want a formal language where we can write things like this, but in a more general setting than just groups.

### 11.3 Syntax of first order logic

**Def:** A **signature**,  $\sigma$ , consists of three disjoint sets:

1. A set of **constant symbols**,  $C_\sigma$
2. A set of **function symbols**,  $F_\sigma$
3. A set of **relation symbols**,  $R_\sigma$

together with a function  $\text{arity}_\sigma : F_\sigma \cup R_\sigma \rightarrow \mathbb{N}_{>0}$ .

For a function symbol  $f$  or relation symbol  $R$ ,  $\text{arity}(f)$  and  $\text{arity}(R)$  denote how many inputs  $f$  and  $R$  have, respectively.

**Example** ( $\sigma_{\text{groups}}$ ):

$$C_{\sigma_{\text{groups}}} = \{e\}, \quad F_{\sigma_{\text{groups}}} = \{\cdot, ^{-1}\}, \quad R_{\sigma_{\text{groups}}} = \emptyset$$

$\text{arity}(\cdot) = 2$ ,  $\text{arity}(^{-1}) = 1$ . Typically written as  $(e, \cdot, ^{-1})$ .

For the rest of these slides, fix a signature  $\sigma$ .

**Def:** The **alphabet** of first order logic over  $\sigma$  consists of all of the following:

1. The constant, function and relation symbols in  $\sigma$ .
2. A countable set of **variable symbols**,  $\{x_0, x_1, x_2, \dots\}$ .
3. The **equality symbol**,  $=$ .
4. The **logical symbols**:  $\exists, \forall, \wedge, \vee, \neg, \rightarrow, \top, \perp$ .
5. The **non-logical symbols**:  $(, ), ,$  (comma).

Note that the variable symbols and equality symbol were not in the alphabet of propositional logic.

**Comment:** We always assume there are no repetitions in the list above. E.g.,  $($  is not a constant, function, or relation symbol.

### 11.3.1 Terms

**Def:** The set of **terms over**  $\sigma$ , denoted  $\text{Terms}(\sigma)$ , is the smallest set of strings over the alphabet of first order logic over  $\sigma$  such that:

1. Every variable symbol is in  $\text{Terms}(\sigma)$ , i.e., for each  $n$ ,  $\langle x_n \rangle \in \text{Terms}(\sigma)$ .
2. Every constant symbol of  $\sigma$  is in  $\text{Terms}(\sigma)$ , i.e., for each  $c \in C_\sigma$ ,  $\langle c \rangle \in \text{Terms}(\sigma)$ .
3. For every function symbol  $f \in F_\sigma$  of arity  $n$  and all terms  $t_1, \dots, t_n \in \text{Terms}(\sigma)$ , the string  $f(t_1, \dots, t_n) \in \text{Terms}(\sigma)$ .  
Technically  $\langle f, ( \rangle + t_1 + \langle , \rangle + t_2 + \dots + \langle , \rangle + t_n + \langle ) \rangle$ .

**Example:** Let  $\sigma$  be the signature with:

- Constant symbols  $a$  and  $b$  ( $C_\sigma = \{a, b\}$ )
- Function symbols  $f, g, h$  with arities 1, 2, 3
- Relation symbols  $R, S$  with arities 2, 3 ( $R_\sigma = \{R, S\}$ )

Some terms:

1.  $a$
2.  $x_{35}$
3.  $f(a)$
4.  $g(f(f(a)), f(x_2))$
5.  $h(f(b), g(x_1, f(a)), f(f(f(x_1))))$

Some non-terms:

1.  $f(a, b)$  ( $f$  has arity 1, not 2)
2.  $S(x_1, x_2, x_3)$  (Relation symbols do not occur in terms)
3.  $g(a \ b)$  (No comma between arguments of  $g$ )

### 11.3.2 Atomic formulas

**Def:** The set of **atomic formulas over**  $\sigma$  is the set of strings over the alphabet of first order logic consisting of:

1.  $\top$  and  $\perp$  (technically  $\langle \top \rangle$  and  $\langle \perp \rangle$ ).
2. For any  $t_1, t_2 \in \text{Terms}(\sigma)$ , the string  $t_1 = t_2$  (technically  $t_1 + \langle = \rangle + t_2$ ).
3. For any relation symbol  $R$  in  $\sigma$  of arity  $n$  and any  $t_1, \dots, t_n \in \text{Terms}(\sigma)$ , the string  $R(t_1, \dots, t_n)$  (technically  $\langle R, \langle \rangle + t_1 + \langle , \rangle + t_2 + \dots + \langle , \rangle + t_n + \langle \rangle \rangle$ ).

**E.g.** With  $\sigma$  as on the previous page:

1.  $\top$
2.  $\perp$
3.  $x_1 = a$
4.  $f(f(a)) = g(x_1, f(b))$
5.  $R(a, b)$
6.  $S(f(x_2), a, g(a, b))$

### 11.3.3 Formulas (a.k.a. well-formed formulas over $\sigma$ )

**Def:** The set of **formulas over**  $\sigma$ , denoted  $\mathcal{L}(\sigma)$ , is the smallest set of strings over the alphabet of first order logic over  $\sigma$  such that:

1. Every atomic formula over  $\sigma$  is in  $\mathcal{L}(\sigma)$ .
2. For every  $\varphi \in \mathcal{L}(\sigma)$  and every variable symbol  $x_i$ ,  $\forall x_i \varphi$  and  $\exists x_i \varphi$  are in  $\mathcal{L}(\sigma)$ .
3. For every  $\varphi, \psi \in \mathcal{L}(\sigma)$ ,  $(\varphi \wedge \psi)$ ,  $(\varphi \vee \psi)$ ,  $(\varphi \rightarrow \psi)$ ,  $(\neg \varphi) \in \mathcal{L}(\sigma)$ .

**E.g.** With  $\sigma$  as on the previous page last:

1.  $\top$
2.  $a = f(b)$
3.  $\forall x_1 (x_1 = a \vee R(x_1, x_2))$

$$4. (\exists x_1 f(x_1) = x_1 \rightarrow (R(a, a) \wedge \forall x_2 S(x_2, a, b)))$$

**Some comments:**

1. In a formula of the form  $\forall x_i \varphi$  or  $\exists x_i \varphi$ , the variable  $x_i$  does not always have to occur in  $\varphi$ .  
E.g.,  $\forall x_2 (a = b \vee R(a, b))$  is fine. So is  $\exists x_1 (a = b \rightarrow b = a)$ .  
But in practice, it's better to avoid doing these.
2. It is possible for  $\exists x_i$  or  $\forall x_i$  to occur multiple times in the same formula.  
E.g.,  $\forall x_1 (R(x_1, a) \vee \forall x_1 S(x_1, x_1, x_1))$  is okay. So is  $\forall x_1 \exists x_1 R(x_1, x_1)$ .
3. We will often write formulas in ways that violate our official definition if doing so improves how easy they are to read.  
E.g., Technically correct:  $\forall x_1 x_1 = x_1$ .  
Easier to read:  $\forall x_1 (x_1 = x_1)$ .

## 11.4 Induction and recursion

All of the basic facts that we proved about the language of propositional logic have analogues for first order logic — with very similar proofs.

In particular:

1. **Structural induction** for terms and for formulas.  
E.g., if you want to prove some property holds of all terms in  $\text{Terms}(\sigma)$ , it is enough to show it holds for variables and constants and that if it holds for  $t_1, \dots, t_n$  it also holds for  $f(t_1, \dots, t_n)$ .
2. **Unique readability** for terms and for formulas.
3. **Structural recursion** for terms and for formulas.

## 12. Semantics of First Order Logic

### 12.1 Introduction

**Goal:** Define semantics of first order logic.

**Semantics** = What formulas and terms mean.

**Comment:** First order logic = logic of mathematical structures. What a formula  $\varphi$  means depends on what structure we interpret  $\varphi$  in. Just like what a propositional formula means depends on how you interpret the propositional symbols.

**Warning:** The definitions in these subsections may look complicated or intimidating. But once you have understood them you will see that they are defining something fairly simple.

### 12.2 Structures

For the rest of these sections, fix a signature  $\sigma$ .

**Def:** A  $\sigma$ -**structure**,  $m$ , consists of:

- A set  $M$  called the **domain**, **universe** or **underlying set** of  $m$ , denoted  $|m|$ .
- For each constant symbol  $c$  of  $\sigma$ , an element  $c^m \in M$ .
- For each function symbol  $f$  of  $\sigma$ , a function  $f^m : M^n \rightarrow M$  where  $n = \text{arity}(f)$ .
- For each relation symbol  $R$  of  $\sigma$ , a set  $R^m \subseteq M^n$  where  $n = \text{arity}(R)$ .

**Idea:** Given a  $\sigma$ -structure  $m$ , we can interpret what terms & formulas over  $\sigma$  mean in  $m$ .

## 12.3 Examples

Throughout these slides we will consider a few running examples.

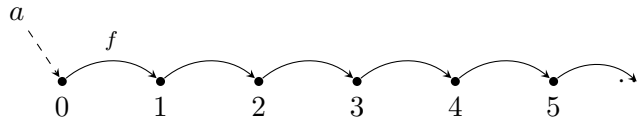
**Example 1:** Let  $\sigma_1$  be the signature with  $C_{\sigma_1} = \{a\}$ ,  $F_{\sigma_1} = \{f\}$ , and  $R_{\sigma_1} = \emptyset$  with  $\text{arity}(f) = 1$ . Sometimes written as  $\sigma_1 = (a, f)$ .

(a) Let  $m_1$  be the  $\sigma_1$ -structure where

$$|m_1| = \mathbb{N}, \quad a^{m_1} = 0, \quad f^{m_1} = (x \mapsto x + 1).$$

Sometimes denoted  $m_1 = (\mathbb{N}, 0, x \mapsto x + 1)$ .

**Picture of  $m_1$ :**

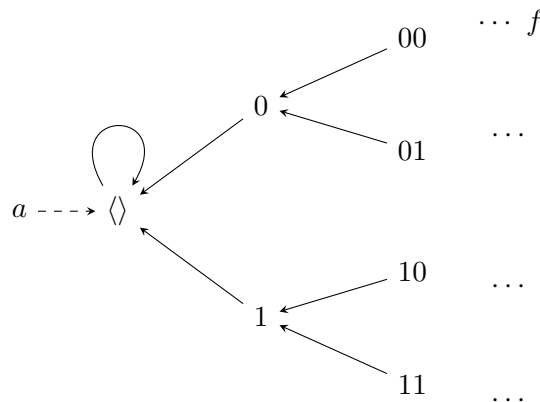


(b) Let  $n_1$  be the  $\sigma_1$ -structure where

$$|n_1| = \text{set of binary strings}, \quad a^{n_1} = \langle \rangle \text{ (empty string)},$$

$$f^{n_1}(x) = \begin{cases} x \text{ with the last char. removed} & \text{if } x \neq \langle \rangle \\ \langle \rangle & \text{if } x = \langle \rangle \end{cases}$$

**Picture of  $n_1$ :**



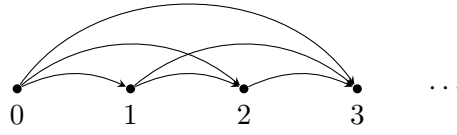


**Example 2:** Let  $\sigma_2$  be the signature with  $C_{\sigma_2} = \emptyset$ ,  $F_{\sigma_2} = \emptyset$ , and  $R_{\sigma_2} = \{S\}$  with  $\text{arity}(S) = 2$ .

(a) Let  $m_2$  be the  $\sigma_2$ -structure where

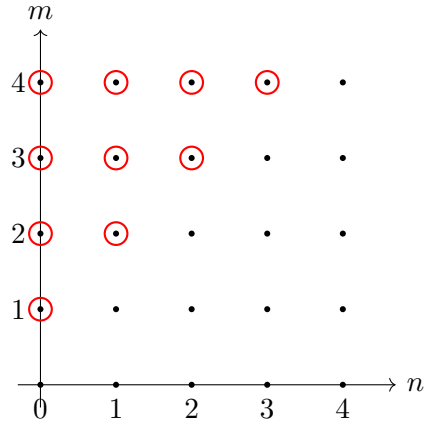
$$|m_2| = \mathbb{N}, \quad S^{m_2} = \{(n, m) \mid n < m\}.$$

**Picture of  $m_2$ :**



**Comment:** An arrow from  $n$  to  $m$  means  $(n, m) \in S^{m_2}$ .

**Another picture:**

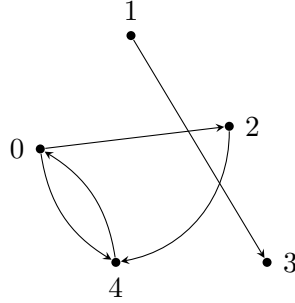


$(n, m)$  is circled  $\iff (n, m) \in S^{m_2}$ .

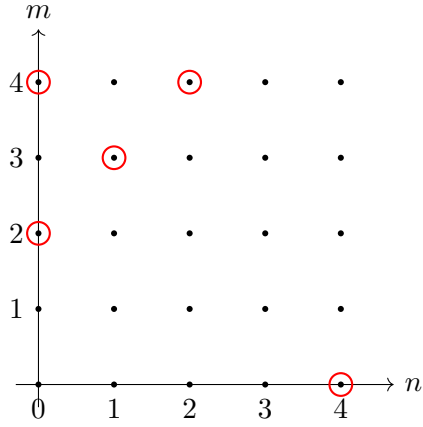
(b) Let  $n_2$  be the  $\sigma_2$ -structure where

$$|n_2| = \{0, 1, 2, 3, 4\}, \quad S^{n_2} = \{(0, 2), (0, 4), (2, 4), (4, 0), (1, 3)\}.$$

**Picture of  $n_2$ :**



Another picture:



## 12.4 Interpreting terms

For the rest of these sections, fix a  $\sigma$ -structure  $m$  with domain  $|m| = M$ .

**Def:** A **variable assignment** for  $m$  is a function  $v : \{x_0, x_1, \dots\} \rightarrow M$ .

**Def:** Given a variable assignment  $v$  for  $M$ , define a function  $\bar{v} : \text{Terms}(\sigma) \rightarrow M$  by structural recursion as follows:

1. For any  $i$ ,  $\bar{v}(x_i) = v(x_i)$ .
2. For any constant symbol  $c$ ,  $\bar{v}(c) = c^m$ .
3. For any function symbol  $f$  of arity  $n$  and terms  $t_1, \dots, t_n$ ,

$$\bar{v}(f(t_1, \dots, t_n)) = f^m(\bar{v}(t_1), \dots, \bar{v}(t_n)).$$

Note that  $\bar{v}$  depends on  $m$ , but this is hidden in the notation.

### 12.4.1 Examples

(a) Let  $m_1$  be the  $\sigma_1$ -structure where

$$|m_1| = \mathbb{N}, \quad a^{m_1} = 0, \quad f^{m_1} = (x \mapsto x + 1).$$

Let  $v : \{x_0, x_1, \dots\} \rightarrow \mathbb{N}$  be defined by  $v(x_n) = n$ .

- $\bar{v}(a) = a^{m_1} = 0$
- $\bar{v}(x_5) = v(x_5) = 5$
- $\bar{v}(f(x_5)) = f^{m_1}(v(x_5)) = f^{m_1}(5) = 5 + 1 = 6$
- $\bar{v}(f(f(f(a)))) = f^{m_1}(f^{m_1}(f^{m_1}(a^{m_1}))) = ((0 + 1) + 1) + 1 = 3$

(b) Let  $n_1$  be the  $\sigma_1$ -structure where

$$|n_1| = \text{set of binary strings}, \quad a^{n_1} = \langle \rangle$$

$$f^{n_1}(x) = \begin{cases} x \text{ with the last char. removed} & \text{if } x \neq \langle \rangle \\ \langle \rangle & \text{if } x = \langle \rangle \end{cases}$$

Let  $v : \{x_0, x_1, \dots\} \rightarrow |n_1|$  be defined by  $v(x_n) = \underbrace{11 \cdots 1}_n$ .

- $\bar{v}(a) = a^{n_1} = \langle \rangle$
- $\bar{v}(x_5) = v(x_5) = 11111$
- $\bar{v}(f(x_5)) = f^{n_1}(v(x_5)) = f^{n_1}(11111) = 1111$
- $\bar{v}(f(f(f(x_5)))) = f^{n_1}(f^{n_1}(f^{n_1}(v(x_5)))) = f^{n_1}(f^{n_1}(f^{n_1}(11111))) = 11$

### 12.5 Interpreting formulas

**Goal:** Define a relation,  $\models$ , called the **satisfaction relation** such that, for any  $\sigma$ -structure  $m$ , variable assignment  $v$  for  $m$ , and formula  $\varphi \in \mathcal{L}(\sigma)$ :

- $(m, v) \models \varphi$  intuitively means  $\varphi$  is true in  $m$  when variables are interpreted according to  $v$ .  
“( $m, v$ ) satisfies  $\varphi$ ”
- $(m, v) \not\models \varphi$  intuitively means  $\varphi$  is false in  $m$  when variables are interpreted according to  $v$ .  
“( $m, v$ ) does not satisfy  $\varphi$ ”

We will define  $(m, v) \models \varphi$  in two steps:

1. First we will define it just for atomic formulas.
2. Then we will use structural induction to define it for all formulas.

### 12.5.1 Atomic formulas

**Def:** For a  $\sigma$ -structure  $m$ , variable assignment  $v$  for  $m$ , and atomic formula  $\varphi$ , define the relation  $(m, v) \models \varphi$  as follows:

1.  $(m, v) \models \top$  and  $(m, v) \not\models \perp$ .
2. If  $\varphi$  has the form  $t_1 = t_2$  for terms  $t_1, t_2$ :

$$(m, v) \models \varphi \iff \bar{v}(t_1) = \bar{v}(t_2).$$

3. If  $\varphi$  has the form  $R(t_1, \dots, t_n)$  for a relation symbol  $R$  and terms  $t_1, \dots, t_n$  then

$$(m, v) \models \varphi \iff (\bar{v}(t_1), \dots, \bar{v}(t_n)) \in R^m.$$

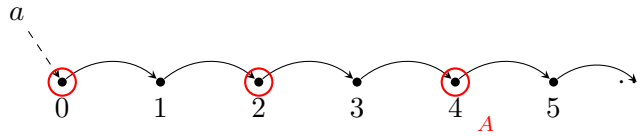
Also,  $(m, v) \not\models \varphi$  if and only if it is not the case that  $(m, v) \models \varphi$ .

**Examples:** Let  $\sigma$  be the signature consisting of  $C_\sigma = \{a\}$ ,  $F_\sigma = \{f\}$  with  $\text{arity}(f) = 1$ , and  $R_\sigma = \{A\}$  with  $\text{arity}(A) = 1$ .

① Let  $m$  be the  $\sigma$ -structure with

$$|m| = \mathbb{N}, \quad a^m = 0, \quad f^m = (x \mapsto x + 1), \quad A^m = \{n \in \mathbb{N} \mid n \text{ is even}\}.$$

**Picture of  $m$ :**



Define  $v$  by  $v(x_n) = n$ .

- (a)  $(m, v) \models f(f(a)) = f(x_1)$ ?

**Yes.**  $\bar{v}(f(f(a))) = (0 + 1) + 1 = 2$ ,  $\bar{v}(f(x_1)) = 1 + 1 = 2$ .

- (b)  $(m, v) \models A(f(f(x_5)))$ ?

**No.**  $\bar{v}(f(f(x_5))) = (5 + 1) + 1 = 7$ , and 7 is not even, so  $7 \notin A^m$ .

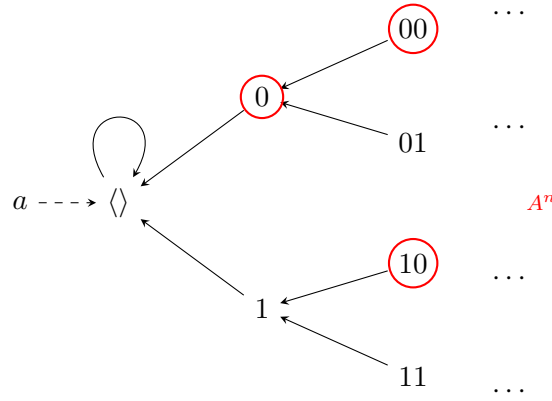
② Let  $n$  be the  $\sigma$ -structure with

$|n| = \text{set of binary strings, } a^n = \langle \rangle,$

$$f^n(x) = \begin{cases} x \text{ with the last char. removed} & \text{if } x \neq \langle \rangle \\ \langle \rangle & \text{if } x = \langle \rangle \end{cases}$$

$$A^n = \{x \in |n| \mid x \neq \langle \rangle \text{ and the last character of } x \text{ is } 0\}.$$

**Picture of  $n$ :**



Define  $v$  by  $v(x_n) = \underbrace{11 \cdots 1}_n$ .

- (a)  $(n, v) \models f(f(a)) = f(x_1)$ ?

**Yes.**  $\bar{v}(f(f(a))) = \langle \rangle = \bar{v}(f(x_1))$ .

- (b)  $(n, v) \models A(f(f(x_5)))$ ?

**No.**  $\bar{v}(f(f(x_5))) = 111$ , and 111 ends with 1, so  $111 \notin A^n$ .

### 12.5.2 All formulas

**Def:** For a  $\sigma$ -structure  $m$ , define by structural recursion on the set of formulas over  $\sigma$ , relations  $(m, v) \models \varphi$  and  $(m, v) \not\models \varphi$ , where  $v$  is any variable assignment for  $m$  and  $\varphi \in \mathcal{L}(\sigma)$ :

1. If  $\varphi$  is atomic then  $(m, v) \models \varphi$  and  $(m, v) \not\models \varphi$  were defined previously.
2. If  $\varphi$  has the form...
  - $\varphi = (\psi_1 \wedge \psi_2)$ :  $(m, v) \models \varphi$  iff  $(m, v) \models \psi_1$  and  $(m, v) \models \psi_2$ .
  - $\varphi = (\psi_1 \vee \psi_2)$ :  $(m, v) \models \varphi$  iff  $(m, v) \models \psi_1$  or  $(m, v) \models \psi_2$ .
  - $\varphi = (\psi_1 \rightarrow \psi_2)$ :  $(m, v) \models \varphi$  iff  $(m, v) \not\models \psi_1$  or  $(m, v) \models \psi_2$ .
  - $\varphi = (\neg\psi)$ :  $(m, v) \models \varphi$  iff  $(m, v) \not\models \psi$ .
3. If  $\varphi = \forall x_i \psi$  then  $(m, v) \models \varphi$  iff for all  $a \in |m|$ ,  $(m, v[x_i/a]) \models \psi$ .
4. If  $\varphi = \exists x_i \psi$  then  $(m, v) \models \varphi$  iff for some  $a \in |m|$ ,  $(m, v[x_i/a]) \models \psi$ .

**Technical definition** needed for  $(m, v) \models \forall x_i \psi$  and  $(m, v) \models \exists x_i \psi$ :

**Def:** Given

- a  $\sigma$ -structure  $m$ ,
- a variable assignment  $v$  for  $m$ ,
- a variable  $x_i$ ,
- and an element  $a \in |m|$ ,

define a variable assignment  $v[x_i/a]$  for  $m$  by

$$(v[x_i/a])(x_j) = \begin{cases} v(x_j) & \text{if } j \neq i \\ a & \text{if } j = i \end{cases}$$

i.e.,  $v[x_i/a]$  is identical to  $v$  except  $x_i$  is assigned to  $a$ .

**Example:**  $|m| = \mathbb{N}$ ,  $v(x_n) = n$ .

|       | $v$ | $v[x_2/17]$ |
|-------|-----|-------------|
| $x_0$ | 0   | 0           |
| $x_1$ | 1   | 1           |
| $x_2$ | 2   | 17          |
| $x_3$ | 3   | 3           |
| $x_4$ | 4   | 4           |

**Examples (continued):** Let  $\sigma$  be the signature consisting of  $C_\sigma = \{a\}$ ,  $F_\sigma = \{f\}$ ,  $R_\sigma = \{A\}$ ,  $\text{arity}(f) = 1$ ,  $\text{arity}(A) = 1$ .

① Let  $m$  be the  $\sigma$  – *signature* with  $|m| = \mathbb{N}$ ,  $a^m = 0$ ,  $f^m = (x \mapsto x + 1)$ ,  $A^m = \{n \in \mathbb{N} \mid n \text{ even}\}$ . Define  $v$  by  $v(x_n) = n$ .

(a)  $(m, v) \models \exists x_1 f(x_1) = x_6$ ?

**Yes.**  $(m, v[x_1/5]) \models f(x_1) = x_6$

(b)  $(m, v) \models \forall x_1 \exists x_2 (f(x_2) = x_1 \wedge A(x_2))$ ?

**No.**  $(m, v[x_1/0]) \not\models \exists x_2 (f(x_2) = x_1 \wedge A(x_2))$ :

if  $x_1$  is assigned to 0, this is false no matter how  $x_2$  is assigned (no  $n \in \mathbb{N}$  satisfies  $n + 1 = 0$ ).

② Let  $n$  be the  $\sigma$  – *signature* with  $|n| = \text{binary strings}$ ,  $a^n = \langle \rangle$ ,

$$f^n(x) = \begin{cases} x \text{ with the last char. removed} & \text{if } x \neq \langle \rangle \\ \langle \rangle & \text{if } x = \langle \rangle \end{cases}$$

$A^n = \{x \in |n| \mid x \neq \langle \rangle \text{ and the last character of } x \text{ is } 0\}$

Define  $v$  by  $v(x_n) = \underbrace{11 \cdots 1}_n$ .

(a)  $(n, v) \models \exists x_1 f(x_1) = x_6$ ?

**Yes.**  $(n, v[x_1/1111110]) \models f(x_1) = x_6$

(b)  $(n, v) \models \forall x_1 \exists x_2 (f(x_2) = x_1 \wedge A(x_2))$ ?

**Yes.** For any  $s \in |n|$ ,

$(n, v[x_1/s][x_2/s + \langle 0 \rangle]) \models f(x_2) = x_1$ ,  $(n, v[x_1/s][x_2/s + \langle 0 \rangle]) \models A(x_2)$ .

## 13. Loose Ends

### 13.1 Free variables

Each variable that appears in a formula  $\varphi$  is classified as either a **free variable** or a **bound variable**.

**Intuitively:**  $x_i$  is a free variable of  $\varphi$  if it appears somewhere in  $\varphi$  not in the scope of any quantifier.

**E.g.**  $\varphi = R(x_1, x_2) \vee \exists x_1 \forall x_3 S(x_1, x_3, x_4)$

- Free variables:  $x_1, x_2, x_4$
- Bound variables:  $x_3$

Note that  $x_1$  is free despite being quantified over because it appears at least once w/out quantifiers.

**Formally:** Define a function  $\text{FreeVar} : \mathcal{L}(\sigma) \rightarrow \mathcal{P}(\{x_0, x_1, x_2, \dots\})$  by structural recursion as follows:

- $\text{FreeVar}(\varphi) = \{x_i \mid x_i \text{ appears in } \varphi\}$  if  $\varphi$  is atomic
- $\text{FreeVar}(\varphi \wedge \psi) = \dots = \text{FreeVar}(\varphi \rightarrow \psi) = \text{FreeVar}(\varphi) \cup \text{FreeVar}(\psi)$
- $\text{FreeVar}(\exists x_i \varphi) = \text{FreeVar}(\forall x_i \varphi) = \text{FreeVar}(\varphi) \setminus \{x_i\}$
- $\text{FreeVar}(\neg \varphi) = \text{FreeVar}(\varphi)$

The point of introducing free variables:

**Thm 13.1:** Suppose  $\varphi$  is a formula,  $m$  is a  $\sigma$ -structure, and  $u, v$  are variable assignments for  $m$  which agree on all free variables of  $\varphi$ . Then

$$(m, v) \models \varphi \iff (m, u) \models \varphi.$$

I.e., when checking if  $(m, v) \models \varphi$ , we only need to know what values  $v$  assigns to the free variables of  $\varphi$ .

**Example:** To know if  $(m, v) \models \forall x_1 f(x_1) = x_2$ , we only need to know  $v(x_2)$  and not  $v(x_1)$  or  $v(x_3), v(x_4), \dots$ .

The proof is a straightforward structural induction. We will skip it.

### 13.1.1 Sentences

**Def:** A formula  $\varphi$  is a **sentence** if it has no free variables, i.e.,  $\text{FreeVar}(\varphi) = \emptyset$ .

**Examples:**

- $\forall x_1 (\exists x_2 R(x_1, x_2) \wedge \exists x_3 \exists x_2 S(x_2, x_3, x_1))$  (sentence)
- $R(x_1, x_2) \vee \exists x_1 \forall x_3 S(x_1, x_3, x_4)$  (not a sentence)



**Cor:** (of Thm 13.1) If  $\varphi$  is a sentence then for all variable assignments  $u$  and  $v$ ,

$$(m, v) \models \varphi \iff (m, u) \models \varphi.$$

i.e.,  $(m, v) \models \varphi$  does not depend on  $v$  at all.

**Notation:** If  $\varphi$  is a sentence, we write  $m \models \varphi$  to mean that for any  $v$ ,  $(m, v) \models \varphi$ . (“ $m$  models  $\varphi$ ” or “ $m$  satisfies  $\varphi$ ”.)

## 13.2 Empty structures are not allowed

When we defined “ $\sigma$ -structure” we did not specify whether the domain of the structure was allowed to be empty.

From now on, we will **not** allow structures with empty domain. I.e., we change the definition of “ $\sigma$ -structure” to require that for any  $\sigma$ -structure  $m$ ,  $|m| \neq \emptyset$ .

**Why do this:** It makes a lot of definitions simpler.

**E.g.,** when  $\varphi$  is a sentence, we defined  $m \models \varphi$  to mean that for any variable assignment  $v$  for  $m$ ,  $(m, v) \models \varphi$ . But if  $|m| = \emptyset$  then there are no variable assignments for  $m$ , so this would mean  $m \models \varphi$  for all sentences  $\varphi$ .

We could get around this by adding a special case to the def. when  $|m| = \emptyset$ , but instead we will just forbid  $|m| = \emptyset$ .

## 13.3 Some conventions

① Suppose  $\text{FreeVar}(\varphi) \subseteq \{x_0, \dots, x_n\}$ . Then we write

$$m \models \varphi(a_0, \dots, a_n)$$

to mean that for any variable assignment  $v$  for  $m$  such that  $v(x_0) = a_0, \dots, v(x_n) = a_n$ , we have  $(m, v) \models \varphi$ .

② We often write  $t \neq s$  to mean  $(\neg t = s)$ .

③ We will write certain symbols using infix notation. E.g., if  $+$  is a binary function symbol, we will usually write  $t + s$  instead of  $+(t, s)$ .

④ We will not follow the formal definition of the language of first order logic very closely & will instead use parentheses, etc., as needed for readability.

E.g.,

- $\exists x (x = y)$  instead of  $\exists x x = y$ .
- $x = y \wedge y = z$  instead of  $(x = y \wedge y = z)$

### 13.4 A non-mathematical example

First order logic can also be used as a model of logical reasoning outside of math. Here's one toy example.

Let  $\sigma$  be the signature with two unary function symbols,  $F$  and  $M$ .

Let  $n$  be the  $\sigma$ -structure with

$|n|$  = all humans in history,  $F^n(x)$  =  $x$ 's father,  $M^n(x)$  =  $x$ 's mother.

- $\exists x \exists y (x \neq y \wedge M(x) = M(y))$   
means *there are 2 people with the same mother*.
- $\forall x \exists y ((M(x) = M(y) \wedge F(x) \neq F(y)) \vee (M(x) \neq M(y) \wedge F(x) = F(y)))$   
means *everyone has a half-sibling*.

### 13.5 Another example

A lot of structures studied in mathematics can be thought of as structures in first-order logic.

**One example:** Let  $\sigma_{NT} = (\text{zero}, \text{one}, \text{add}, \text{mul}, \text{LT})$  where:

- zero, one are constants
- add, mul are binary functions
- LT is a binary relation

$\mathcal{L}(\sigma_{NT})$  is called the “**language of number theory**”.

Let  $n$  be the  $\sigma_{NT}$ -structure with:

$$|n| = \mathbb{N}, \quad \text{zero}^n = 0, \quad \text{one}^n = 1,$$

$$\text{add}^n(a, b) = a + b, \quad \text{mul}^n(a, b) = ab, \quad \text{LT}^n = \{(a, b) \in \mathbb{N}^2 \mid a < b\}.$$

**Some interesting formulas:**

$$\textcircled{1} \varphi_{\text{even}}(x) = \exists y (\text{add}(y, y) = x).$$

$$n \models \varphi_{\text{even}}(n) \iff n \text{ is even.}$$

$$\textcircled{2} \varphi_{\text{prime}}(x) = x \neq \text{zero} \wedge x \neq \text{one} \wedge \forall y \forall z (\text{mul}(y, z) = x \rightarrow (y = x \vee z = x)).$$

$$n \models \varphi_{\text{prime}}(n) \iff n \text{ is prime.}$$

$$\textcircled{3} \forall x (\text{LT}(\text{add}(\text{one}, \text{one}), x) \wedge \varphi_{\text{even}}(x) \rightarrow \exists u \exists v (\varphi_{\text{prime}}(u) \wedge \varphi_{\text{prime}}(v) \wedge \text{add}(u, v) = x))$$

(Goldbach's conjecture)

**Comment:** We used one, zero, add, mul, LT to keep the symbols in the language distinct from  $0, 1 \in \mathbb{N}$ ,  $+: \mathbb{N}^2 \rightarrow \mathbb{N}$ , etc. Often this is not done & people write  $\sigma_{NT} = (0, 1, +, \times, <)$ .

## 14. Definable Sets and Related Concepts

### 14.1 Logical implication and logical equivalence

Fix a signature  $\sigma$ .

**Def:** Given a set of  $\mathcal{L}(\sigma)$ -formulas,  $\Gamma$ , a  $\sigma$ -structure  $m$ , and a variable assignment  $v$  for  $m$ ,  $(m, v)$  **satisfies**  $\Gamma$ , written  $(m, v) \models \Gamma$ , if for every  $\varphi \in \Gamma$ ,  $(m, v) \models \varphi$ .

**Def:** Given a set of  $\mathcal{L}(\sigma)$ -formulas,  $\Gamma$ , and an  $\mathcal{L}(\sigma)$ -formula  $\varphi$ ,  $\Gamma$  **logically implies**  $\varphi$ , written  $\Gamma \models \varphi$ , if for all  $\sigma$ -structures  $m$  and variable assignments  $v$  for  $m$ , if  $(m, v) \models \Gamma$  then  $(m, v) \models \varphi$ .

**Notation:**  $\varphi \models \psi$  means  $\{\varphi\} \models \psi$ .

**Def:**  $\mathcal{L}(\sigma)$ -formulas  $\varphi$  and  $\psi$  are **logically equivalent**, written  $\varphi \equiv \psi$ , if  $\varphi \models \psi$  and  $\psi \models \varphi$ .

**Example:** Let  $\sigma$  be the signature with one binary relation symbol  $S$ .

$$\textcircled{1} \exists x \forall y S(x, y) \models \forall y \exists x S(x, y).$$

To see why, fix  $m$  and  $v$  such that  $(m, v) \models \exists x \forall y S(x, y)$ .

We know:  $(m, v) \models \exists x \forall y S(x, y)$

$\Rightarrow$  for some  $a \in |m|$ ,  $(m, v[x/a]) \models \forall y S(x, y)$

$\Rightarrow$  for any  $b \in |m|$ ,  $(m, v[x/a][y/b]) \models S(x, y)$

We want:  $(m, v) \models \forall y \exists x S(x, y)$ .

ETS: For any  $b \in |m|$ ,  $(m, v[y/b]) \models \exists x S(x, y)$ .

ETS: For any  $b \in |m|$ ,  $(m, v[y/b][x/a]) \models S(x, y)$ . (same  $a$  as above)

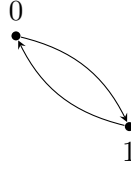
This follows from the line above because  $v[x/a][y/b] = v[y/b][x/a]$ .

②  $\forall y \exists x S(x, y) \not\models \exists x \forall y S(x, y)$ .

To see this, it's enough to find one  $\sigma$ -structure  $m$  such that  $m \models \forall y \exists x S(x, y)$  and  $m \not\models \exists x \forall y S(x, y)$ .

Let  $|m| = \{0, 1\}$  and  $S^m = \{(0, 1), (1, 0)\}$ .

**Picture of  $m$ :**



$m \models \forall y \exists x S(x, y)$ :

- $(m, v[y/0][x/1]) \models S(x, y)$
- $(m, v[y/1][x/0]) \models S(x, y)$

$m \not\models \exists x \forall y S(x, y)$ :

- $(m, v[x/0][y/0]) \not\models S(x, y)$
- $(m, v[x/1][y/1]) \not\models S(x, y)$

**Some logical equivalences(De Morgan's laws):**

1.  $\neg \forall x \varphi(x) \equiv \exists x \neg \varphi(x)$
2.  $\neg \exists x \varphi(x) \equiv \forall x \neg \varphi(x)$

$$3. \neg(\varphi \vee \psi) \equiv \neg\varphi \wedge \neg\psi$$

$$4. \neg(\varphi \wedge \psi) \equiv \neg\varphi \vee \neg\psi$$

### 14.1.1 Notation

Just like with propositional logic, the symbol  $\models$  is used in two fairly different ways (each of which has several variations).

#### Use #1

$$(m, v) \models \varphi$$

#### Use #2

$$\Gamma \models \varphi$$

#### Variations:

$$m \models \varphi \quad (\varphi \text{ is a sentence})$$

$$\varphi \models \psi \quad (\{\varphi\} \models \psi)$$

$$m \models \varphi(a_0, \dots, a_n)$$

$$\Gamma \models \Delta$$

$$(\text{FreeVar}(\varphi) \subseteq \{x_0, \dots, x_n\})$$

$$(\text{For all } \varphi \in \Delta, \Gamma \models \varphi)$$

$$(m, v) \models \Gamma \quad (\Gamma \text{ is a set of formulas}) \quad \varphi \equiv \psi \quad (\varphi \models \psi \text{ and } \psi \models \varphi)$$

$$m \models \Gamma \quad (\Gamma \text{ is a set of sentences}) \quad \Gamma \equiv \Delta \quad (\Gamma \models \Delta \text{ and } \Delta \models \Gamma)$$

### 14.1.2 Tautologies

**Def:** An  $\mathcal{L}(\sigma)$ -formula  $\varphi$  is:

- a **tautology** if for every  $\sigma$ -structure  $m$  and variable assignment  $v$  for  $m$ ,  $(m, v) \models \varphi$ .
- **satisfiable** if for some  $m$  and  $v$ ,  $(m, v) \models \varphi$ .
- a **contradiction** if for all  $m$  and  $v$ ,  $(m, v) \not\models \varphi$ .

**Example:** Some tautologies:

$$1. f(c) \neq c \rightarrow \exists x \exists y (x \neq y)$$

$$2. \forall x \exists y (S(x, y) \vee \forall z \neg S(x, z))$$

## 14.2 Theories and models

Fix a signature  $\sigma$ .

**Def:** Given a  $\sigma$ -structure  $m$ , the **theory of  $m$** , denoted  $\text{Th}(m)$ , is the set of sentences that  $m$  satisfies. I.e.,

$$\text{Th}(m) = \{\varphi \in \mathcal{L}(\sigma) \mid \varphi \text{ is a sentence and } m \models \varphi\}.$$

**Def:** Given a set of  $\mathcal{L}(\sigma)$ -sentences  $\Gamma$ , the **class of models of  $\Gamma$** , denoted  $\text{Mod}(\Gamma)$ , is the set of  $\sigma$ -structures that satisfy  $\Gamma$ . I.e.,

$$\text{Mod}(\Gamma) = \{m \mid m \models \Gamma\}.$$

**Def:** A class  $A$  of  $\sigma$ -structures is a:

- **elementary class** if  $A = \text{Mod}(\Gamma)$  for some  $\Gamma$  (also called *axiomatizable*)
- **basic elementary class** if  $A = \text{Mod}(\{\varphi\})$  for some  $\varphi$  (also called *finitely axiomatizable*)

**Example:** Recall  $\sigma_{\text{groups}} = (\cdot, {}^{-1}, e)$  (binary function, unary function, constant).

The class of all groups is a basic elementary class (i.e., finitely axiomatizable):

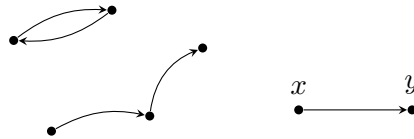
$$\text{Class of groups} = \text{Mod} \left( \left\{ \begin{array}{l} \forall x \forall y \forall z (x \cdot y) \cdot z = x \cdot (y \cdot z) \\ \wedge \forall x (x \cdot e = x \wedge e \cdot x = x) \\ \wedge \forall x (x^{-1} \cdot x = e \wedge x \cdot x^{-1} = e) \end{array} \right\} \right).$$

**Comment:** The class of *finite* groups is **not** an elementary class. Later this semester we will develop the tools necessary to prove this.

**Another example:** Let  $\sigma_{\text{graphs}}$  be the signature with one binary relation  $E$ .

**Def:** A **directed graph** is any  $\sigma_{\text{graphs}}$ -structure.

**Picture:**



A directed arrow from  $x$  to  $y$  denotes  $E(x, y)$ .

**Def:** A **path** in a directed graph  $G$  is a sequence of elements  $a_0, \dots, a_n$  of  $G$  such that for each  $i < n$ ,  $(a_i, a_{i+1}) \in E^G$ .



**Def:** The **diameter** of a directed graph  $G$  is the least  $K \in \mathbb{N}$  s.t. for all  $u, v$  in  $G$  there is a path  $a_0, \dots, a_n$  in  $G$  with  $n \leq K$ ,  $a_0 = u$  and  $a_n = v$ . Or  $\infty$  if no such  $K$  exists.

① The class of directed graphs with diameter  $\leq 5$  is a basic elementary class (i.e., finitely axiomatizable).

Let

$$\varphi = \forall x \forall y \exists z_0 \dots \exists z_5 \left( z_0 = x \wedge z_5 = y \wedge \bigwedge_{i < 5} (E(z_i, z_{i+1}) \vee z_i = z_{i+1}) \right).$$

② The class of directed graphs with finite diameter is **not** an elementary class. (Later in the semester we will be able to prove this.)

③ The class of directed graphs with diameter  $\infty$  is an elementary class but not a basic elem. class. Consider the (infinite) set of formulas:

$$\left\{ \neg \forall x \forall y \exists z_0 \dots \exists z_k \left( z_0 = x \wedge z_k = y \wedge \bigwedge_{i < k} (E(z_i, z_{i+1}) \vee z_i = z_{i+1}) \right) \mid k \in \mathbb{N} \right\}$$

### 14.2.1 Elementary equivalence

**Def:**  $\sigma$ -structures  $m$  and  $n$  are **elementary equivalent**, denoted  $m \equiv n$ , if  $\text{Th}(m) = \text{Th}(n)$ .

Equivalently: for every sentence  $\varphi$ ,  $m \models \varphi \iff n \models \varphi$ .

**Example:**  $(\mathbb{Z}, 0, +) \not\equiv (\mathbb{N}, 0, +)$ .

- $(\mathbb{Z}, 0, +) \models \forall x \exists y (x + y = 0)$
- $(\mathbb{N}, 0, +) \not\models \forall x \exists y (x + y = 0)$

**Comment:** We don't yet have the tools necessary to prove  $m \equiv n$  in any interesting cases. But we will soon.

**Comment:** Given a structure  $m$ , determining  $\text{Th}(m)$  is often difficult or even impossible.

### 14.3 Definable sets

Fix a signature  $\sigma$  and a  $\sigma$ -structure  $m$  with domain  $M$ .

**Def:**  $A \subseteq M^n$  is **definable** if there is an  $\mathcal{L}(\sigma)$ -formula  $\varphi$  with  $\text{FreeVar}(\varphi) \subseteq \{x_0, \dots, x_{n-1}\}$  such that

$$A = \{(a_0, \dots, a_{n-1}) \in M^n \mid m \models \varphi(a_0, \dots, a_{n-1})\}.$$

**Def:**  $A \subseteq M^n$  is **definable with parameters** if there is an  $\mathcal{L}(\sigma)$ -formula  $\varphi$  with  $\text{FreeVar}(\varphi) \subseteq \{x_0, \dots, x_{n+m-1}\}$  and elements  $b_0, \dots, b_{m-1} \in M$  such that

$$A = \{(a_0, \dots, a_{n-1}) \in M^n \mid m \models \varphi(a_0, \dots, a_{n-1}, b_0, \dots, b_{m-1})\}.$$

**Example:** Let  $\sigma$  be the signature with binary function symbols  $+$  and  $\times$ . Let  $(\mathbb{N}, +, \times)$  denote the  $\sigma$ -structure with domain  $\mathbb{N}$  where  $+$  is interpreted as addition and  $\times$  as multiplication.

①  $\{0\}$  is definable.

Let  $\varphi = \forall y (x + y = y)$ . For all  $n \in \mathbb{N}$ ,

$$(\mathbb{N}, +, \times) \models \varphi(n) \iff n = 0.$$

②  $\{n \in \mathbb{N} \mid n \text{ is even}\}$  is definable.

Let  $\varphi = \exists y (x = y + y)$ .

$$(\mathbb{N}, +, \times) \models \varphi(n) \iff n \text{ is even}.$$

③  $\{(n, m) \in \mathbb{N}^2 \mid n < m\}$  is definable.

Let  $\varphi = \exists z (x + z = y \wedge x \neq y)$ .

$$(\mathbb{N}, +, \times) \models \varphi(n, m) \iff n < m.$$



**Example:** Let  $\sigma$  be the signature with one binary relation symbol  $<$ . Let  $(\mathbb{Z}, <)$  be the  $\sigma$ -structure with domain  $\mathbb{Z}$  and  $<$  interpreted as the less-than relation on  $\mathbb{Z}$ .

The set  $\{n \in \mathbb{Z} \mid n \geq 0\}$  is **not** definable. (We will see how to prove this soon.)

But it *is* definable with parameters:

Let  $\varphi = y < x \vee x = y$ . Then

$$\{n \in \mathbb{Z} \mid n \geq 0\} = \{n \in \mathbb{Z} \mid (\mathbb{Z}, <) \models \varphi(n, 0)\}.$$

(0 is being used as a parameter.)

## 15. Morphisms

### 15.1 Introduction

**Goal for this lecture:** Introduce a method to help show that certain structures are elementarily equivalent and that certain sets are not definable.

**Key idea:** Look at functions between structures.

Briefly, we will define a notion of isomorphisms of  $\sigma$ -structures and show:

1. If  $m$  and  $n$  are isomorphic then  $m \equiv n$ .
2. If  $f$  is an isomorphism from  $m$  to  $n$ ,  $A \subseteq M$  is a definable subset of  $m$ , and  $a \in M$ , then  $a \in A \iff f(a) \in A$ .

So to show  $A$  is **not** definable, it is enough to find an isomorphism  $f$  and an  $a \in M$  such that  $a \in A$  and  $f(a) \notin A$ .

### 15.2 Morphisms of $\sigma$ -structures

Fix a signature  $\sigma$ .

**Def:** A **homomorphism** from a  $\sigma$ -structure  $m$  to a  $\sigma$ -structure  $n$  is a function  $f : M \rightarrow N$  such that:

1. For each constant symbol  $c$ ,  $f(c^m) = c^n$ .
2. For each function symbol  $g$  of arity  $n$  and all  $a_1, \dots, a_n \in M$ ,

$$f(g^m(a_1, \dots, a_n)) = g^n(f(a_1), \dots, f(a_n)).$$

3. For each relation symbol  $R$  of arity  $n$  and all  $a_1, \dots, a_n \in M$ ,

$$(a_1, \dots, a_n) \in R^m \Rightarrow (f(a_1), \dots, f(a_n)) \in R^n.$$

Note that condition (3) is not if and only if. Some textbooks have a different definition of homomorphism which requires

$$(a_1, \dots, a_n) \in R^m \iff (f(a_1), \dots, f(a_n)) \in R^n.$$

**Def:** A homomorphism  $f$  from  $m$  to  $n$  is an:

- **embedding** if  $f$  is injective and for all relation symbols  $R$  of arity  $n$  and all  $a_1, \dots, a_n \in M$ ,

$$(a_1, \dots, a_n) \in R^m \iff (f(a_1), \dots, f(a_n)) \in R^n.$$

- **isomorphism** if  $f$  is an embedding which is also surjective (and hence bijective).
- **automorphism** if  $f$  is an isomorphism and  $m = n$ .

**Notation:** If there is an isomorphism from  $m$  to  $n$ , we say  $m$  and  $n$  are **isomorphic** and write  $m \cong n$ .

**Comment:** If  $\sigma$  is the signature of group theory and  $m$  and  $n$  are groups, then a homomorphism from  $m$  to  $n$  is exactly a group homomorphism, and an isomorphism  $m \rightarrow n$  is exactly a group isomorphism.

### 15.2.1 Examples

- ① Let  $\sigma$  be the signature with one binary relation symbol  $R$ .

$$\begin{aligned} m : M = \{1, 2, 3, 4\}, \quad R^m &= \{(1, 3), (3, 1), (2, 1), (2, 3), (3, 4)\}, \\ n : N = \{1, 2, 3\}, \quad R^n &= \{(1, 1), (1, 2), (3, 2)\}. \end{aligned}$$

**Picture of  $m$  and  $n$ :**



- (a) Let  $f : M \rightarrow N$  be defined by  $f(1) = 1$ ,  $f(2) = 1$ ,  $f(3) = 1$ ,  $f(4) = 2$ .

Then  $f$  is a homomorphism, but *not* an embedding: e.g.  $(1, 2) \notin R^m$  but  $(f(1), f(2)) = (1, 1) \in R^n$ .

- (b) Let  $g : M \rightarrow N$  be defined by  $g(1) = 3$ ,  $g(2) = 1$ ,  $g(3) = 2$ ,  $g(4) = 1$ .

Then  $g$  is *not* a homomorphism: e.g.  $(2, 1) \in R^m$  but  $(g(2), g(1)) = (1, 3) \notin R^n$ .

② Let  $\sigma$  be the signature with one binary function symbol  $+$ . Let  $m = (\mathbb{Z}, +)$  (i.e.  $+^m$  is addition)

(a) Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  defined by  $f(n) = 2n$ .

- $f$  is a homomorphism  $m \rightarrow m$ : for any  $n, m \in \mathbb{Z}$ ,

$$f(n + m) = 2(n + m) = 2n + 2m = f(n) + f(m).$$

- $f$  is an embedding: Easy to show  $f$  is injective.
- $f$  is *not* an automorphism:  $f$  is not surjective.

(b) Let  $g : \mathbb{Z} \rightarrow \mathbb{Z}$  defined by  $g(n) = n + 2$ .

Then  $g$  is *not* a homomorphism  $m \rightarrow m$ :

$$g(5 + 3) = g(8) = 10, \quad g(5) + g(3) = 7 + 5 = 12 \quad (\text{not equal}).$$

③ Let  $\sigma$  be the signature with one binary relation symbol  $<$ . Let  $m = (\mathbb{Z}, <)$ . Define  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  by  $f(n) = n + 7$ .

Then  $f$  is an automorphism of  $m$ :

- Easy to check  $f$  is bijective.
- For any  $n, m$ :  $n < m \iff n + 7 < m + 7 \iff f(n) < f(m)$ .

## 15.3 Using morphisms

Why did we introduce morphisms? Two key ideas:

1. If  $m \cong n$  then  $m$  and  $n$  “look the same.”

In particular, for any sentence  $\varphi$ ,  $m \models \varphi \iff n \models \varphi$ .

**Consequence:**  $m \cong n \Rightarrow m \equiv n$ .

2. If  $f : m \rightarrow m$  is an automorphism and  $a \in M$ , then  $a$  and  $f(a)$  “look the same” inside  $M$ . In particular, for any formula  $\varphi$  with one free variable,  $m \models \varphi(a) \iff m \models \varphi(f(a))$ .

**Consequence:** To show  $A \subseteq M$  is not definable in  $m$ , it is enough to find an automorphism  $f$  of  $m$  and an  $a \in M$  such that  $a \in A$  and  $f(a) \notin A$ .

We will now make this more formal.

### 15.3.1 The morphism theorem

Fix a signature  $\sigma$  and  $\sigma$ -structures  $m$  and  $n$ .

**Def:** If  $v$  is a variable assignment for  $m$  and  $f : M \rightarrow N$  is a function,  $v_f$  is the variable assignment for  $n$  defined by

$$v_f(x_i) = f(v(x_i)).$$

(I.e.  $v_f$  is literally just  $f \circ v$ .)

**Thm 15.1:** For any function  $f : M \rightarrow N$  and variable assignment  $v$  for  $m$ :

- If  $f$  is a homomorphism then for any term  $t$ ,  $f(\overline{v}(t)) = \overline{v_f}(t)$ , and for any atomic formula  $\varphi$ ,  $(m, v) \models \varphi \Rightarrow (n, v_f) \models \varphi$ .
- If  $f$  is an embedding then for any formula  $\varphi$  with no quantifiers,  $(m, v) \models \varphi \iff (n, v_f) \models \varphi$ .
- If  $f$  is an isomorphism then for any formula  $\varphi$ ,  $(m, v) \models \varphi \iff (n, v_f) \models \varphi$ .

For an isomorphism,  $m \models \varphi(a_1, \dots, a_n) \iff n \models \varphi(f(a_1), \dots, f(a_n))$ .

### Corollaries of the morphism theorem:

**Cor 15.2:** If  $m$  and  $n$  are isomorphic then  $m \equiv n$  (elementarily equivalent).

*proof:* Let  $f : m \rightarrow n$  be an isomorphism. For any sentence  $\varphi$ ,

$$\begin{aligned} m \models \varphi &\iff \text{for some } v, (m, v) \models \varphi \\ &\iff \text{for some } v, (n, v_f) \models \varphi \\ &\iff n \models \varphi. \end{aligned}$$

**Cor 15.3:** If  $f : m \rightarrow m$  is an automorphism and  $A \subseteq M^n$  is a definable set, then for all  $a_1, \dots, a_n \in M$ ,

$$(a_1, \dots, a_n) \in A \iff (f(a_1), \dots, f(a_n)) \in A.$$

(So to show  $A$  is not definable, it is enough to find  $a_1, \dots, a_n$  such that  $(a_1, \dots, a_n) \in A$  and  $(f(a_1), \dots, f(a_n)) \notin A$ .)

*proof:* Suppose  $A$  is defined by  $\varphi$ . Then

$$\begin{aligned} (a_1, \dots, a_n) \in A &\iff m \models \varphi(a_1, \dots, a_n) \\ &\iff m \models \varphi(f(a_1), \dots, f(a_n)) \\ &\iff (f(a_1), \dots, f(a_n)) \in A. \quad \square \end{aligned}$$

### 15.3.2 Examples

① Let  $\sigma$  be the signature with one binary relation symbol  $<$ , and let  $m = (\mathbb{Z}, <)$ , so  $M = \mathbb{Z}$ .

**Claim:**  $\{n \in \mathbb{Z} \mid n \geq 0\}$  is not definable in  $m$ .

*proof:* Let  $f : \mathbb{Z} \rightarrow \mathbb{Z}$  be defined by  $f(n) = n - 5$ .

Easy to check  $f$  is an automorphism.  $2 \in \{n \in \mathbb{Z} \mid n \geq 0\}$ , but  $f(2) = -3 \notin \{n \in \mathbb{Z} \mid n \geq 0\}$ . By Cor 15.3, this implies  $\{n \in \mathbb{Z} \mid n \geq 0\}$  is not definable.

② Let  $\sigma$  be the signature with one binary function symbol  $g$ . Let  $m$  be the  $\sigma$ -structure with

$$\begin{aligned} |m| &= \{n \in \mathbb{N} \mid n \geq 1 \text{ and the only prime divisors of } n \text{ are } 2 \text{ and } 3\}, \\ &= \{1, 2, 3, 4, 6, 8, \dots\} \end{aligned}$$

$$g^m(n, m) = nm \quad (\text{multiplication}).$$

Let  $n$  be the  $\sigma$ -structure with

$$|n| = \mathbb{Z} \times \mathbb{Z}, \quad g^n((a, b), (c, d)) = (a + c, b + d).$$

**Claim:**  $m \equiv n$ .

*proof:* Define  $f : n \rightarrow m$  by  $f(a, b) = 2^a \cdot 3^b$ . Not hard to show  $f$  is an isomorphism. Hence  $n \cong m$ , so  $n \equiv m$ .

## 15.4 Proof of the morphism theorem

For this entire section, fix a signature  $\sigma$ ,  $\sigma$ -structures  $m$  and  $n$ , and a function  $f : M \rightarrow N$ .

**Thm (terms):** If  $f$  is a homomorphism then for any variable assignment  $v$  for  $m$  and any term  $t$ ,  $f(\bar{v}(t)) = \bar{v}_f(t)$ .

*proof:* By structural induction on  $t$ .

- **Case 1:**  $t = x_i$ .  $f(\bar{v}(t)) = f(v(x_i)) = v_f(x_i) = \bar{v}_f(t)$  (by def. of  $v_f$ ).
- **Case 2:**  $t = c$ , a constant symbol.  $f(\bar{v}(t)) = f(c^m) = c^n = \bar{v}_f(t)$  (by def. of “homomorphism”).
- **Case 3:**  $t = g(s_1, \dots, s_n)$  where  $\forall i \leq n, f(\bar{v}(s_i)) = \bar{v}_f(s_i)$ .

$$\begin{aligned} f(\bar{v}(g(s_1, \dots, s_n))) &= f(g^m(\bar{v}(s_1), \dots, \bar{v}(s_n))) \\ &= g^n(f(\bar{v}(s_1)), \dots, f(\bar{v}(s_n))) \quad (\text{by def. of homomorphism}) \\ &= g^n(\bar{v}_f(s_1), \dots, \bar{v}_f(s_n)) \quad (\text{inductive hypothesis}) \\ &= \bar{v}_f(g(s_1, \dots, s_n)). \end{aligned}$$

**Thm (atomic formulas):** If  $f$  is a homomorphism then for any variable assignment  $v$  for  $m$  and any atomic formula  $\varphi$ , if  $(m, v) \models \varphi$  then  $(n, v_f) \models \varphi$ . Moreover, if  $f$  is an embedding, then  $(m, v) \models \varphi \iff (n, v_f) \models \varphi$ .

*proof:*

- **Case 1:**  $\varphi = \top$  or  $\varphi = \perp$ . Trivial.

- **Case 2:**  $\varphi$  is  $t = s$ .

$$\begin{aligned}
(m, v) \models \varphi &\iff \bar{v}(t) = \bar{v}(s) \\
&\Rightarrow f(\bar{v}(t)) = f(\bar{v}(s)) \\
&\iff \bar{v}_f(t) = \bar{v}_f(s) && \text{(by Thm(terms))} \\
&\iff (n, v_f) \models \varphi.
\end{aligned}$$

If  $f$  is an embedding, then (being injective) the “ $\Rightarrow$ ” step is also a “ $\iff$ ,”.

- **Case 3:**  $\varphi = R(t_1, \dots, t_n)$ .

$$\begin{aligned}
(m, v) \models \varphi &\iff (\bar{v}(t_1), \dots, \bar{v}(t_n)) \in R^m \\
&\Rightarrow (f(\bar{v}(t_1)), \dots, f(\bar{v}(t_n))) \in R^n && \text{(by def. of homomorphism)} \\
&\iff (\bar{v}_f(t_1), \dots, \bar{v}_f(t_n)) \in R^n && \text{(by Thm(terms))} \\
&\iff (n, v_f) \models \varphi.
\end{aligned}$$

If  $f$  is an embedding, the “ $\Rightarrow$ ” step is also a “ $\iff$ ,” so all steps are biconditionals.

**Thm (isomorphisms):** If  $f$  is an isomorphism then for any formula  $\varphi$  and any variable assignment  $v$  for  $m$ ,  $(m, v) \models \varphi \iff (n, v_f) \models \varphi$ .

*proof:* By structural induction on  $\varphi$ .

**Base case:**  $\varphi$  is an atomic formula. This essentially follows from the previous page.

**Inductive case:** Assume  $\varphi$  and  $\psi$  have the desired properties. The cases we need to consider are  $\varphi \wedge \psi$ ,  $\varphi \vee \psi$ ,  $\varphi \rightarrow \psi$ ,  $\neg\varphi$ ,  $\exists x_i \varphi$ ,  $\forall x_i \varphi$ . We will just handle the representative cases below.

- **Case  $\wedge$ :**

$$\begin{aligned}
(m, v) \models \varphi \wedge \psi &\iff (m, v) \models \varphi \text{ and } (m, v) \models \psi \\
&\iff (n, v_f) \models \varphi \text{ and } (n, v_f) \models \psi && \text{(by IH)} \\
&\iff (n, v_f) \models \varphi \wedge \psi.
\end{aligned}$$

- **Case  $\exists$ :**  $\varphi = \exists x_i \psi$ .



( $\Rightarrow$ ) Suppose  $(m, v) \models \exists x_i \psi$ . Then for some  $a \in M$ ,  $(m, v[x_i/a]) \models \psi$ .  
 By IH,

$$(n, (v[x_i/a])_f) \models \psi \Rightarrow (n, v_f[x_i/f(a)]) \models \psi \Rightarrow (n, v_f) \models \exists x_i \psi,$$

using the identity  $(v[x_i/a])_f = v_f[x_i/f(a)]$ .

( $\Leftarrow$ ) Suppose  $(n, v_f) \models \exists x_i \psi$ . Then for some  $b \in N$ ,  $(n, v_f[x_i/b]) \models \psi$ .  
 Since  $f$  is surjective,  $f(a) = b$  for some  $a \in M$ . Then

$$(n, v_f[x_i/f(a)]) \models \psi$$

$$\Rightarrow (n, (v[x_i/a])_f) \models \psi \Rightarrow (m, v[x_i/a]) \models \psi \text{ (by IH)} \Rightarrow (m, v) \models \exists x_i \psi.$$

• **Case  $\forall$ :**  $\varphi = \forall x_i \psi$ .

We prove the contrapositive of each direction, i.e. that  $(m, v) \not\models \forall x_i \psi \Rightarrow (n, v_f) \not\models \forall x_i \psi$ .

( $\Rightarrow$ ) Suppose  $(m, v) \not\models \forall x_i \psi$ . Then for some  $a \in M$ ,  $(m, v[x_i/a]) \not\models \psi$ .  
 Since  $f$  is surjective,  $f(a) = b$  for some  $a \in M$ . Then

$$(n, v_f[x_i/f(a)]) \not\models \psi$$

$$\Rightarrow (n, (v[x_i/a])_f) \not\models \psi \Rightarrow (m, v[x_i/a]) \not\models \psi \text{ (by IH)} \Rightarrow (m, v) \not\models \forall x_i \psi.$$

( $\Leftarrow$ ) Suppose  $(n, v_f) \not\models \forall x_i \psi$ . Then for some  $a \in N$ ,  $(n, v_f[x_i/a]) \not\models \psi$ .  
 Then

$$(m, v[x_i/a]) \not\models \psi$$

$$\Rightarrow (n, (v[x_i/a])_f) \not\models \psi \text{ (by IH)} \Rightarrow (n, v_f[x_i/f(a)]) \not\models \psi \Rightarrow (n, v_f) \not\models \forall x_i \psi.$$

## 16. The Back-and-forth Method

### 16.1 Introduction

**Previously:** Isomorphisms are useful for showing that structures are elementarily equivalent or that sets are not definable.

But so far we don't have any systematic way to construct isomorphisms.

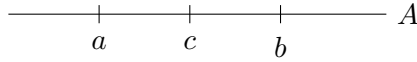
**This lecture:** An example of one useful method for constructing isomorphisms — the “back-and-forth” method.

Instead of describing the method in general, we will see how it works in one specific case.

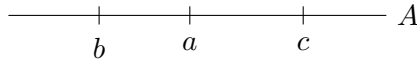
### 16.2 Dense linear orders

**Def:** A **dense linear order** (or **DLO**, for short) is a linear order  $(A, <_A)$  such that:

1. Between any two elements there is another element: for any  $a, b \in A$ , if  $a <_A b$  then there is some  $c \in A$  such that  $a <_A c <_A b$ .



2. There is no smallest or largest element: for any  $a \in A$  there are  $b, c \in A$  such that  $b <_A a$  and  $a <_A c$ .



**Comment:** Sometimes “dense linear order” just means a linear order satisfying (1), and a linear order satisfying (1) & (2) is called a “dense linear order without endpoints.”

**Comment:** The class of DLOs is finitely axiomatizable:

$$\text{DLOs} = \text{Mod} \left( \left\{ \begin{array}{l} \forall x \neg(x < x), \\ \forall x \forall y \forall z (x < y \wedge y < z \rightarrow x < z), \\ \forall x \forall y (x < y \vee y < x \vee y = x), \\ \forall x \forall y (x < y \rightarrow \exists z (x < z \wedge z < y)), \\ \forall x \exists y \exists z (y < x \wedge x < z) \end{array} \right\} \right).$$

The first three sentences are the axioms for linear orders; the fourth expresses density; the fifth expresses having no endpoints.

**Examples:**

- ①  $\mathbb{Z}$  — not a DLO. Nothing between 0 and 1.
- ②  $\mathbb{R}$  — DLO.
- ③  $[0, 1)$  — not a DLO. 0 is the smallest element.
- ④  $(0, 1)$  — DLO.
- ⑤  $\mathbb{Q}$  — DLO.

Using the back-and-forth method, we will prove:

**Thm 16.1 (Cantor):** Any two **countable** DLOs are isomorphic. (Yes, that Cantor.)

### 16.3 Countability

**Def:** A set  $X$  is **countable** if either  $X = \emptyset$  or there is a surjective function  $f : \mathbb{N} \rightarrow X$  (possibly with repetitions).

Think of  $f$  as an “enumeration” of  $X$ . I.e. let  $x_0 = f(0), x_1 = f(1), x_2 = f(2), \dots$ . Then  $X = \{x_0, x_1, x_2, \dots\}$ .

**Examples:**

- ①  $\mathbb{Z}$  is countable:  $\mathbb{Z} = \{0, -1, 1, -2, 2, \dots\}$ . Formally, define  $f : \mathbb{N} \rightarrow \mathbb{Z}$  by

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ -(n+1)/2 & \text{if } n \text{ is odd.} \end{cases}$$

- ②  $\mathbb{Q}$  is countable. List all fractions  $\frac{p}{q}$  in an array (rows indexed by the denominator  $q \geq 1$ , columns by the numerator  $p \in \{0, -1, 1, -2, 2, \dots\}$ ) and enumerate them along the diagonals (skipping repetitions and undefined entries):

$$\mathbb{Q} = \{0, -1, 1, 0, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{3}, 1, -2, \dots\}.$$

|          | 0             | -1             | 1             | -2             | 2             | ... |
|----------|---------------|----------------|---------------|----------------|---------------|-----|
| 1        | $\frac{0}{1}$ | $\frac{-1}{1}$ | $\frac{1}{1}$ | $\frac{-2}{1}$ | $\frac{2}{1}$ | ... |
| 2        | $\frac{0}{2}$ | $\frac{-1}{2}$ | $\frac{1}{2}$ | $\frac{-2}{2}$ | $\frac{2}{2}$ | ... |
| 3        | $\frac{0}{3}$ | $\frac{-1}{3}$ | $\frac{1}{3}$ | $\frac{-2}{3}$ | $\frac{2}{3}$ | ... |
| $\vdots$ |               |                |               |                |               |     |

**Thm (Cantor):**  $\mathbb{R}$  is not countable.

## 16.4 Back-and-forth

**Thm 16.1 (Cantor):** Any two countable DLOs are isomorphic.

**Idea:** Suppose  $(A, <_A)$  and  $(B, <_B)$  are countable DLOs. Let

$$A = \{a_0, a_1, \dots\}, \quad B = \{b_0, b_1, \dots\}$$

(such enumerations exist because  $A$  and  $B$  are countable). Build a bijection  $f : A \rightarrow B$  one step at a time. On step  $n$ :

- Assign  $f(a_n) = b_k$  for some  $k$  (to make sure  $a_n \in \text{dom}(f)$ );
- Assign  $f(a_\ell) = b_n$  for some  $\ell$  (to make sure  $b_n \in \text{range}(f)$ , so that  $f$  is surjective).

Along the way, we need to ensure:

1.  $f$  stays injective;
2.  $f$  is order-preserving, i.e.  $a_i <_A a_j \Rightarrow f(a_i) <_B f(a_j)$ .

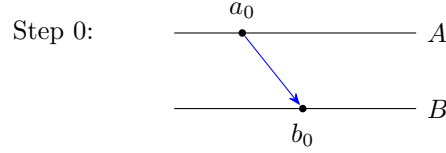
The DLO property ensures we can always maintain these.

A little more precisely, on each step:

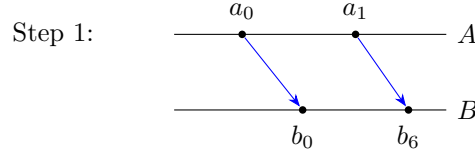
1. If  $a_n$  is not yet in the domain of  $f$ , find some  $b_k$  that we can map  $a_n$  to while keeping  $f$  injective & order-preserving, and assign  $f(a_n) = b_k$ . (The fact that such a  $b_k$  exists follows from  $B$  being a DLO.)

2. If  $b_n$  is not yet in the range of  $f$ , find some  $a_\ell$  that we can map to  $b_n$  while keeping  $f$  injective & order-preserving, and assign  $f(a_\ell) = b_n$ . (The fact that such an  $a_\ell$  exists follows from  $A$  being a DLO.)

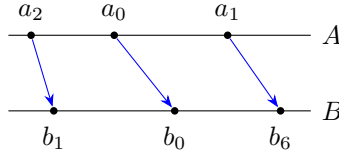
**Illustration of the first few steps** (an arrow  $a_i \rightarrow b_j$  means  $f(a_i) = b_j$ ):



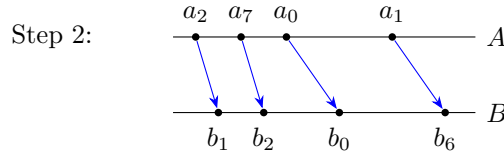
On Step 0 we assign  $f(a_0) = b_0$ .



On Step 1 we assign  $f(a_1) = b_6$  (say); since  $a_0 <_A a_1$ , to keep  $f$  order-preserving we need  $f(a_0) <_B f(a_1)$ .



To put  $b_1$  into the range we assign  $f(a_2) = b_1$ ; since  $b_1 <_B b_0$ , we need  $a_2 <_A a_0$ .



On Step 2 we don't need to assign  $f(a_2)$  (already assigned on the previous step); to put  $b_2$  into the range we assign  $f(a_7) = b_2$ . And so on.

### 16.4.1 A formal proof

**Thm 16.1 (Cantor):** Any two countable DLOs are isomorphic.

*proof:* Let  $(A, <_A)$  and  $(B, <_B)$  be DLOs with  $A = \{a_0, a_1, a_2, \dots\}$  and  $B = \{b_0, b_1, b_2, \dots\}$ . We will build a sequence of functions  $p_0, p_1, p_2, \dots$  such that:

1. For each  $n$ ,  $\text{dom}(p_n) \subseteq A$  and  $\text{range}(p_n) \subseteq B$ .
2. For each  $n$ ,  $\text{dom}(p_n)$  is finite.
3. For each  $n$ ,  $a_n \in \text{dom}(p_n)$  and  $b_n \in \text{range}(p_n)$ .
4. For each  $n$ ,  $p_{n+1}$  extends  $p_n$ . I.e.  $\text{dom}(p_n) \subseteq \text{dom}(p_{n+1})$  and for all  $a \in \text{dom}(p_n)$ ,  $p_n(a) = p_{n+1}(a)$ .
5. For each  $n$ ,  $p_n$  is injective & order-preserving.

Intuitively,  $p_n$  is what  $f$  looks like after  $n$  steps in the informal description above.

Given such a sequence, define  $f : A \rightarrow B$  by

$$f(a) = \text{the unique element } b \in B \text{ such that } p_n(a) = b \text{ for some } p_n.$$

Then:

- Property (3) ensures  $\text{dom}(f)$  is all of  $A$ .
- Property (4) ensures  $f$  is well-defined.
- Property (3) ensures  $f$  is surjective.
- Properties (4) & (5) ensure  $f$  is injective & order-preserving.

**Claim:** There is some sequence  $p_0, p_1, p_2, \dots$  with the properties listed above.

*proof of claim:* Define  $p_0, p_1, p_2, \dots$  inductively.

**Base case:** Let  $p_0$  be the function with  $\text{dom}(p_0) = \{a_0\}$  defined by  $p_0(a_0) = b_0$ .

**Inductive case:** Assume  $p_0, \dots, p_n$  have the listed properties; we will define  $p_{n+1}$ . We define  $p_{n+1}$  in two steps: on the first step we add  $a_{n+1}$  to the domain (producing an intermediate function  $q_{n+1}$ ); on the second step we use  $q_{n+1}$  to define  $p_{n+1}$ , adding  $b_{n+1}$  to the range.

**Defining  $p_{n+1}$ , Step 1 (define  $q_{n+1}$ ):**

**Case 1:**  $a_{n+1} \in \text{dom}(p_n)$ . In this case let  $q_{n+1} = p_n$ .

**Case 2:**  $a_{n+1} \notin \text{dom}(p_n)$ . Let  $\text{dom}(p_n) = \{x_0, x_1, \dots, x_m\}$  where  $x_0 <_A x_1 <_A \dots <_A x_m$  (possible since  $\text{dom}(p_n)$  is finite).

- **Case 2a:**  $a_{n+1} <_A x_0$ . Let  $k$  be least such that  $b_k \notin \text{range}(p_n)$  and  $b_k <_B p_n(x_0)$  (such  $b_k$  exists because  $B$  has no smallest element). Let  $q_{n+1}$  be the function with  $\text{dom}(q_{n+1}) = \text{dom}(p_n) \cup \{a_{n+1}\}$  defined by

$$q_{n+1}(x) = \begin{cases} p_n(x) & \text{if } x \neq a_{n+1}, \\ b_k & \text{if } x = a_{n+1}. \end{cases}$$

- **Case 2b:**  $x_m <_A a_{n+1}$ . Similar to Case 2a (such  $b_k$  exists because  $B$  has no largest element).
- **Case 2c:**  $x_0 <_A a_{n+1} <_A x_m$ . Let  $i$  be such that  $x_i <_A a_{n+1} <_A x_{i+1}$ . Let  $k$  be least such that  $b_k \notin \text{range}(p_n)$  and  $p_n(x_i) <_B b_k <_B p_n(x_{i+1})$  (such  $b_k$  exists because  $B$  is a DLO). Let  $q_{n+1}$  be the function with  $\text{dom}(q_{n+1}) = \text{dom}(p_n) \cup \{a_{n+1}\}$  defined by

$$q_{n+1}(x) = \begin{cases} p_n(x) & \text{if } x \neq a_{n+1}, \\ b_k & \text{if } x = a_{n+1}. \end{cases}$$

**Defining  $p_{n+1}$ , Step 2 (define  $p_{n+1}$  from  $q_{n+1}$ ):**

Note that since  $q_{n+1}$  is order-preserving, writing  $\text{dom}(q_{n+1}) = \{x_0, x_1, \dots, x_m\}$  with  $x_0 <_A x_1 <_A \dots <_A x_m$ , we have  $q_{n+1}(x_0) <_B q_{n+1}(x_1) <_B \dots <_B q_{n+1}(x_m)$ .

**Case 1:**  $b_{n+1} \in \text{range}(q_{n+1})$ . Let  $p_{n+1} = q_{n+1}$ .

**Case 2:**  $b_{n+1} \notin \text{range}(q_{n+1})$ .

- **Case 2a:**  $b_{n+1} <_B q_{n+1}(x_0)$ . Let  $\ell$  be least such that  $a_\ell <_A x_0$  (such  $a_\ell$  exists because  $A$  has no smallest element). Let  $p_{n+1}$  be the function with  $\text{dom}(p_{n+1}) = \text{dom}(q_{n+1}) \cup \{a_\ell\}$  defined by

$$p_{n+1}(x) = \begin{cases} q_{n+1}(x) & \text{if } x \in \text{dom}(q_{n+1}), \\ b_{n+1} & \text{if } x = a_\ell. \end{cases}$$

- **Case 2b:**  $q_{n+1}(x_m) <_B b_{n+1}$ . Similar to Case 2a (such  $a_\ell$  exists because  $A$  has no largest element).
- **Case 2c:**  $q_{n+1}(x_0) <_B b_{n+1} <_B q_{n+1}(x_m)$ . Let  $i$  be such that  $q_{n+1}(x_i) <_B b_{n+1} <_B q_{n+1}(x_{i+1})$ . Let  $\ell$  be least such that  $x_i <_A a_\ell <_A x_{i+1}$  (such  $a_\ell$  exists because  $A$  is a DLO). Let  $p_{n+1}$  be the function with  $\text{dom}(p_{n+1}) = \text{dom}(q_{n+1}) \cup \{a_\ell\}$  defined by

$$p_{n+1}(x) = \begin{cases} q_{n+1}(x) & \text{if } x \in \text{dom}(q_{n+1}), \\ b_{n+1} & \text{if } x = a_\ell. \end{cases}$$

One checks in each case that  $p_{n+1}$  has properties (1)–(5), completing the induction.

## 17. Elementary Substructures

### 17.1 Introduction

**Goal for the next 2–3 lectures:** Develop a method to show that structures are elementarily equivalent even when they are not isomorphic.

**Key tool:** Löwenheim–Skolem theorem.

In order to explain what this theorem says, we need two new concepts:

1. substructures
2. elementary substructures.

A substructure is a generalization of concepts like subgroup, subring, subgraph, etc.

### 17.2 Substructures

Fix a signature  $\sigma$  and a  $\sigma$ -structure  $m$ .

**Def:** A **substructure** of  $m$  is a  $\sigma$ -structure  $n$  such that:

1.  $|n| \subseteq |m|$ .
2. For each constant symbol  $c$ ,  $c^n = c^m$ .



3. For each function symbol  $f$  of arity  $n$  and all  $a_1, \dots, a_n \in |n|$ ,  $f^n(a_1, \dots, a_n) = f^m(a_1, \dots, a_n)$ .
4. For each relation symbol  $R$  of arity  $n$ ,  $R^n = R^m \cap |n|^n$ .

Equivalently, the inclusion map  $|n| \hookrightarrow |m|$  is an embedding.

**Examples:** Let  $\sigma$  be a signature with one constant, one binary function, and one binary relation. Let  $m = (\mathbb{Z}, 0, +, <)$ .

- ①  $(\mathbb{N}, 0, +, <)$  — substructure.
- ②  $(\mathbb{N}, 0, \times, <)$  — not a substructure.
- ③  $(\mathbb{Z}, 0, +, <)$  — substructure.
- ④  $(\mathbb{Z}, 0, +, \leq)$  — not a substructure.
- ⑤  $(\mathbb{Q}, 0, +, <)$  — not a substructure.
- ⑥  $(2\mathbb{Z}, 0, +, <)$  — substructure.

### 17.2.1 Induced substructures

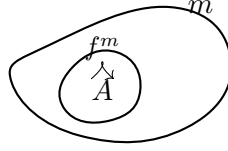
Substructures are usually defined by taking a subset of  $|m|$  and copying  $m$ 's interpretations of the symbols.

**More precisely:** Suppose  $A \subseteq |m|$  and that:

1. For each constant symbol  $c$ ,  $c^m \in A$ .
2. For each function symbol  $f$  of arity  $n$  and all  $a_1, \dots, a_n \in A$ ,  $f^m(a_1, \dots, a_n) \in A$ .

(I.e.  $A$  contains each  $c^m$  and is closed under  $f^m$  for each  $f$  in  $\sigma$ .) Then we say that  $A$  **induces a substructure of  $m$** , and the **substructure induced by  $A$**  is the  $\sigma$ -structure  $n$  defined by:

- $|n| = A$ ;
- for each constant  $c$ ,  $c^n = c^m$ ;
- for each function  $f$ ,  $f^n(a_1, \dots, a_n) = f^m(a_1, \dots, a_n)$ ;
- for each relation  $R$ ,  $R^n = A^n \cap R^m$ .



**Examples:**

- ①  $\mathbb{N}$  induces a substructure of  $(\mathbb{Z}, +)$ .    Yes.
- ②  $\{0\}$  induces a substructure of  $(\mathbb{Z}, +)$ .    Yes.
- ③  $\{1\}$  induces a substructure of  $(\mathbb{Z}, +)$ .    No ( $1 + 1 \notin \{1\}$ ).

**Comment:** If  $G$  is a group, then  $A \subseteq G$  induces a substructure of  $G$  if and only if  $A$  is a subgroup of  $G$ . If  $R$  is a ring, then  $A \subseteq R$  induces a substructure of  $R$  if and only if  $A$  is a subring of  $R$ .

### 17.3 Preserving formulas

Substructures preserve quantifier-free formulas. More formally:

**Thm 17.1:** Suppose  $n$  is a substructure of  $m$ . Then for every formula  $\varphi$  which does not contain any quantifiers and for all  $a_1, \dots, a_n \in |n|$ ,

$$n \models \varphi(a_1, \dots, a_n) \iff m \models \varphi(a_1, \dots, a_n).$$

(This can also be seen as a special case of the theorem about embeddings.)

This can be proved using structural induction. The key point is that  $m$  and  $n$  interpret terms in the same way. We will skip the proof.

### 17.4 Elementary substructures

There is a special name for substructures that preserve all first-order formulas.

**Def:** A substructure  $n \subseteq m$  is an **elementary substructure** of  $m$ , written  $n \preceq m$ , if for all formulas  $\varphi$  and all  $a_1, \dots, a_n \in |n|$ ,

$$n \models \varphi(a_1, \dots, a_n) \iff m \models \varphi(a_1, \dots, a_n).$$

All substructures preserve quantifier-free formulas. Only elementary substructures preserve all formulas.

**Example:**  $(\mathbb{N}, <)$  is a substructure of  $(\mathbb{Z}, <)$ , but not an elementary substructure:

$$(\mathbb{N}, <) \models \exists x \forall y (\neg y < x), \quad (\mathbb{Z}, <) \not\models \exists x \forall y (\neg y < x).$$

**Comment:** If  $n \preccurlyeq m$  we also say  $m$  is an **elementary extension** of  $n$ .

**Comment (important point):** If  $n$  is a substructure of  $m$ , saying that  $n$  is an elementary substructure of  $m$  is stronger than saying  $n \equiv m$ .

Roughly,  $n \equiv m$  means  $n$  and  $m$  agree on sentences, while  $n \preccurlyeq m$  means they agree on all formulas.

**Example:** Let  $n = (2\mathbb{Z}, <)$  (where  $2\mathbb{Z}$  is the set of even integers) and  $m = (\mathbb{Z}, <)$ .

**Claim:**  $n \equiv m$ .

*proof:*  $f : 2\mathbb{Z} \rightarrow \mathbb{Z}$  defined by  $f(a) = a/2$  is an isomorphism  $n \rightarrow m$ .

**Claim:**  $n \not\preccurlyeq m$ .

*proof:* Let  $\varphi := \exists z (x < z \wedge z < y)$ . Then  $n \not\models \varphi(0, 2)$  but  $m \models \varphi(0, 2)$ .

## 17.5 Elementary substructures and automorphisms

**Comment:** Proving that  $n \preccurlyeq m$  is usually not that easy (which partly explains why we have not yet given a nontrivial example).

We will now introduce one method for showing that  $n \preccurlyeq m$ , which involves constructing isomorphisms.

**Recall:** If  $f$  is an isomorphism from  $n$  to  $m$ , then for any formula  $\varphi$  and  $a_1, \dots, a_n$  in  $n$ ,

$$n \models \varphi(a_1, \dots, a_n) \iff m \models \varphi(f(a_1), \dots, f(a_n)).$$

**Cor 17.2:** Suppose  $n$  is a substructure of  $m$  such that for all  $a_1, \dots, a_n$  in  $n$  there is an isomorphism  $f$  from  $n$  to  $m$  such that  $f(a_1) = a_1, \dots, f(a_n) = a_n$ . Then  $n \preccurlyeq m$ .

*proof:* Fix  $\varphi$  and  $a_1, \dots, a_n \in |n|$ . Let  $f$  be as above. We have:

$$n \models \varphi(a_1, \dots, a_n) \xLeftrightarrow[f \text{ is an iso.}] m \models \varphi(f(a_1), \dots, f(a_n)) \xLeftrightarrow[f(a_i)=a_i] m \models \varphi(a_1, \dots, a_n).$$

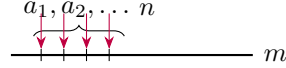
**Example:** Let  $m = (\mathbb{Q}, <)$  and  $n = (\mathbb{Q} \cap (0, 1), <)$ .

**Claim:**  $n \preccurlyeq m$ .

*proof:* Fix  $a_1 < a_2 < \dots < a_n$  in  $\mathbb{Q} \cap (0, 1)$ . It is enough to find an isomorphism from  $n$  to  $m$  that fixes  $a_1, \dots, a_n$ . Such an automorphism  $f : \mathbb{Q} \cap (0, 1) \rightarrow \mathbb{Q}$  is

$$f(x) = \begin{cases} a_1 + \frac{1}{a_1} - \frac{1}{x} & \text{if } 0 < x < a_1, \\ x & \text{if } a_1 \leq x \leq a_n, \\ a_n - \frac{1}{1-a_n} + \frac{1}{1-x} & \text{if } a_n < x < 1. \end{cases}$$

One checks this is an automorphism.



**Comment:** Such an automorphism can be constructed more easily (but less concretely) using the back-and-forth method.

## 18. The Löwenheim–Skolem Theorem

### 18.1 The Löwenheim–Skolem theorem

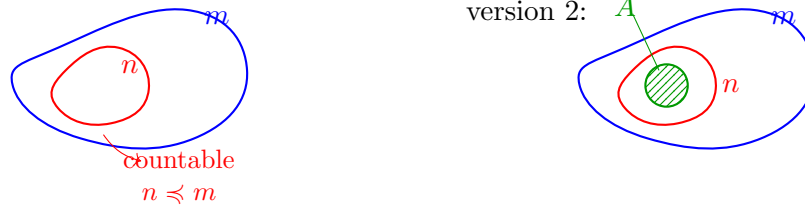
The Löwenheim–Skolem theorem is a tool for constructing elementary substructures. It is also useful for proving elementary equivalence.

**Thm 18.1 (Löwenheim–Skolem thm):** Every infinite structure in a countable signature has a countable elementary substructure.

(Here “countable signature” means there are only countably many constant, function, and relation symbols.) The proof actually gives a stronger (but more complicated) statement, which is sometimes useful:

**Thm 18.2 (Löwenheim–Skolem, version 2):** If  $m$  is an infinite structure in a countable signature, then for every countable set  $A \subseteq |m|$  there is a countable elementary substructure of  $m$  which contains  $A$ .

**Picture:**



### 18.2 Using the Löwenheim–Skolem theorem

The Löwenheim–Skolem theorem is sometimes useful for showing non-isomorphic structures are elementarily equivalent.

**Thm 18.3:** Any two DLOs are elementarily equivalent.

*proof:* Let  $m_1$  and  $m_2$  be DLOs.

$$\begin{aligned}
 \text{L-S thm} &\Rightarrow \exists n_1 \preccurlyeq m_1, n_2 \preccurlyeq m_2 \text{ with } |n_1|, |n_2| \text{ countable} \\
 n_1 \preccurlyeq m_1, n_2 \preccurlyeq m_2 &\Rightarrow n_1, n_2 \text{ are also DLOs} \\
 n_1, n_2 \text{ countable DLOs} &\Rightarrow n_1 \cong n_2 \Rightarrow n_1 \equiv n_2 \\
 m_1 \equiv n_1 \equiv n_2 \equiv m_2 &\Rightarrow m_1 \equiv m_2.
 \end{aligned}$$

In particular,  $(\mathbb{Q}, <) \equiv (\mathbb{R}, <)$ .